Possibilistic Variational Inference

Learning the Precision of the Gaussian Variational Candidate

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Internship Context

- April to August 2026, jointly at
  - LPSM, CNRS and Sorbonne Université, Paris, 1.5 months (background phase)
  - SPMS, Nanyang Technological University, Singapore, 3 months (reproduction, landscapes, contribution & report writing)
- Supervised by Badr-Eddine Chérief-Abdellatif (LPSM) and Jérémie Houssineau (NTU)

The paper this internship extends

J. Singh, S. Wongso, J. Houssineau, B.-E. Chérief-Abdellatif. *Maxitive Donsker–Varadhan Formulation for Possibilistic Variational Inference*. arXiv:2511.21223, 2026 [1].

What this internship addresses

Singh et al. [1]: **expectation** learned, but **precision** fixed by hand or tuned heuristically, never derived from the variational objectives themselves.

Outline

Possibilistic Variational Inference and Its Limits

Learning the Precision of a Gaussian Possibilistic Candidate

SAM Application, Algorithms, and Experimental Validation

Discussion and Conclusion

Possibilistic Variational Inference and Its Limits

Epistemic Uncertainty and Probability Dilution

- **Epistemic uncertainty**: lack of knowledge about a fixed, deterministic $\theta^\star$, with no repeatable experiment to draw from.
- **Improper priors**: $\pi(\theta) \propto 1$ on unbounded Θ is not integrable, and Jeffreys' prior $\pi_J \propto \sqrt{\det I(\theta)}$ can be improper.
- **False confidence** [2]: as $\sigma \to \infty$, $\mathbb{P}(\theta^\star \in A) \to 0$ even when $\mu \in A$ always.
- **Key intuition**: possibility theory [3, 4], a framework representing complete ignorance without diluting local plausibility.

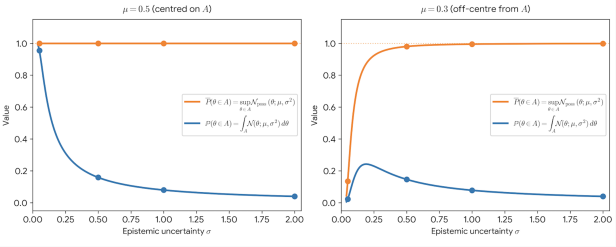


Figure. Probability dilution for a target set $A = [0.4, 0.6]$, at $\mu = 0.5$ and $\mu = 0.3$.

Possibility Functions and Possibilistic Inference

Definition (Possibility function, 5)

$f : \Theta \to [0, 1]$ measurable with $\sup_\theta f(\theta) = 1$. Associated possibility measure: $\bar{P}(A) \coloneqq \sup_{\theta \in A} f(\theta)$. Candidate space $\mathcal{F}(\Theta)$ ($\sup g = 1$) replaces $\mathcal{P}(\Theta)$ ($\int q = 1$), uninformative prior $\pi \equiv 1$ naturally proper.

Possibilistic Bayes' rule, 5

$$g^\star_{\max}(\theta) = \frac{e^{-\ell(\theta)}\pi(\theta)}{Z_{\max}} = \frac{e^{-\hat{\ell}(\theta)}}{Z_{\max}}, \qquad Z_{\max} \coloneqq \sup_{\theta \in \Theta} e^{-\hat{\ell}(\theta)}$$

$\hat{\ell} \coloneqq \ell - \log \pi$. $Z_{\max}$ measures **consistency** between prior and likelihood, not average marginal fit.

Definition (Possibilistic expectation, 5)

When unique, the expectation of $f \in \mathcal{F}(\Theta)$ is its mode, $\mathbb{E}^*[\theta]_f \coloneqq \arg\max_{\theta \in \Theta} f(\theta)$, not an integral mean.

- **Practical**: a single deterministic pass at the mode suffices at test time, no Monte Carlo sampling.
- **Structural**: $Z_{\max}$ already an optimization (sup), turning VI from integral-to-optimization into optimization-to-optimization.

The Maxitive Donsker–Varadhan Formula

Theorem (Maxitive Donsker–Varadhan [1])

Let $\pi : \Theta \to (0, 1]$ and $h \coloneqq -\ell$. Then:

$$\log Z_{\max} \coloneqq \log \sup_{\theta \in \Theta} e^{h(\theta)} \pi(\theta) = \sup_{g \in \mathcal{F}(\Theta)} \inf_{\theta} \left\{ h(\theta) - \log \tfrac{g(\theta)}{\pi(\theta)} \right\} = \inf_{g \in \mathcal{F}(\Theta)} \sup_{\theta} \left\{ h(\theta) - \log \tfrac{g(\theta)}{\pi(\theta)} \right\}$$

- **Probabilistic Donsker–Varadhan analogue** [6] (ν: reference measure, $\int e^h \, \mathrm{d}\nu < \infty$):

  $$\log \int_{\Theta} e^{h(\theta)} \, \nu(\mathrm{d}\theta) = \sup_{\rho \in \mathcal{P}(\Theta)} \left\{ \int h \, \mathrm{d}\rho - \mathrm{KL}(\rho \| \nu) \right\}$$

  recovers the ELBO at $h = -\ell$, $\nu = \pi$: $\mathrm{ELBO}(q) \coloneqq -\mathbb{E}_q[\ell(\theta)] - \mathrm{KL}(q \| \pi)$.
  $\mathrm{KL}(q \| p) \coloneqq \int \log \frac{\mathrm{d}q}{\mathrm{d}p} \, \mathrm{d}q$ if $q \ll p$, $+\infty$ otherwise.

- **Two variational objectives vs. one single ELBO**:
  - Sup-inf yields the **lower bound**: $\underline{\mathrm{CBO}}(g) \coloneqq \inf_\theta \left\{ -\ell(\theta) - \log \frac{g(\theta)}{\pi(\theta)} \right\}$
  - Inf-sup yields the **upper bound**: $\overline{\mathrm{CBO}}(g) \coloneqq \sup_\theta \left\{ -\ell(\theta) - \log \frac{g(\theta)}{\pi(\theta)} \right\}$

LCBO, UCBO, and the Sandwich Decomposition

Exact sandwich identity:

$$\underline{\mathrm{CBO}}(g) + D_{\max}(g \| g^\star_{\max}) = \log Z_{\max} = \overline{\mathrm{CBO}}(g) - D_{\max}(g^\star_{\max} \| g)$$

$$\Longrightarrow \ \underline{\mathrm{CBO}}(g) \le \log Z_{\max} \le \overline{\mathrm{CBO}}(g)$$

Probabilistic analogue (ELBO):

$$\log Z_{\mathrm{add}} = \mathrm{ELBO}(q) + \mathrm{KL}\big(q \,\|\, \pi(\cdot \mid x_{1:n})\big)$$

Max-relative entropy ($D_{\max} \ge 0$, $+\infty$ if $\mathrm{supp}(g) \not\subseteq \mathrm{supp}(f)$):

$$D_{\max}(g \| f) := \sup_{\theta \in \mathrm{supp}(f)} \log \frac{g(\theta)}{f(\theta)}$$

- **Maximising $\underline{\mathbf{CBO}}$**: mode-seeking (under-estimates $g^\star$), $\arg\max_g \underline{\mathrm{CBO}}(g) = \{g \in \mathcal{F} : g \preceq g^\star_{\max}\}$.
- **Minimising $\overline{\mathbf{CBO}}$**: support-covering (over-estimates $g^\star$), $\arg\min_g \overline{\mathrm{CBO}}(g) = \{g \in \mathcal{F} : g^\star_{\max} \preceq g\}$.
- **Intersection**: $g^\star_{\max}$ the unique canonical solution, $\{g^\star_{\max}\} = \arg\max_g \underline{\mathrm{CBO}}(g) \cap \arg\min_g \overline{\mathrm{CBO}}(g)$.

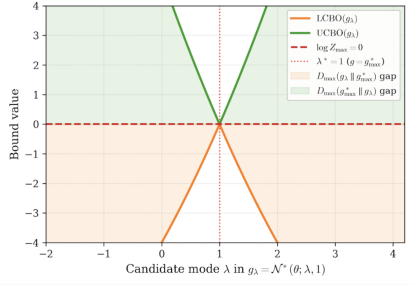


Figure. Sandwich decomposition: lower and upper bounds sandwiching $\log Z_{\max}$.

The CBOpt Updates and the Gap They Leave Open

Signed objective and update [1]

$$\mathrm{CBO}_s(g) := \operatorname*{ext}_{\theta s}\Big\{-\ell(\theta) - \log \tfrac{g(\theta)}{\pi(\theta)}\Big\}, \qquad \eta_{t+1} = \eta_t - \rho_t\, H_s^{-1}\nabla\hat{\ell}(\eta_t) = \eta_t - \rho_t\, s\, \frac{\bar{g}_t + \delta\eta_t}{h_t + \delta - \vartheta_t}$$

$s \in \{+1, -1\}$. Signed curvature $H_s \approx s\ \mathrm{diag}(h_t + \delta - \vartheta_t) \succ 0$. $\bar{g}_t, h_t$: mini-batch gradient and diagonal curvature. δ: L^2 prior precision, division element-wise.

- **Candidate**: restricted to isotropic Gaussian $\mathcal{N}^*(\theta; \eta, \vartheta^{-1} I)$.
- **Curvature tracking**: ensures $H_s \succ 0$ via $\vartheta_t = \gamma c_t$ with scaling factor γ and moving envelope c_t (momentum β_3, shifted curvature $\tilde{h}_t := h_t + \delta$):
  - $s = +1$: $\vartheta_t < \tilde{h}_t \implies \gamma \in [0, 1)$, $c_t = \min\{\beta_3 c_{t-1}, \tilde{h}_t\}$ (`uCBOpt-adapt`)
  - $s = -1$: $\vartheta_t > \tilde{h}_t \implies \gamma > 1$, $c_t = \max\{\beta_3 c_{t-1}, \tilde{h}_t\}$ (`lCBOpt-adapt`)

The gap

In neither case is ϑ derived or optimised from the variational objective itself.

Landscapes on Tractable Posteriors

Four tractable targets with known $g^{\star}_{\max}$ in closed form: Gaussian (sanity check), Bernoulli–Beta, Laplace, bimodal mixture.

Systematic pattern

Both $D_{\max}$ terms collapse to numerical noise wherever the candidate family matches the target's structure. Whenever there is a structural mismatch (support, tails, multimodality), one direction explodes by orders of magnitude.

Order-p check

A softened $D_p \to D_{\max}$ test confirms the ranking is invariant, gaps stem from family misspecification, not from $D_{\max}$ itself.

Left unexplained

Directly maximising $\underline{\mathrm{CBO}}$ over precision collapses ($\vartheta \to \infty$), the phenomenon formalised next.

Learning the Precision of a Gaussian Possibilistic Candidate

Formalising the Collapse

Isotropic candidate $g_{\eta,\vartheta}$, L^2 prior of precision δ, ℓ twice continuously differentiable (almost everywhere for ReLU networks [7]). Now seek to update ϑ at fixed η, letting ϑ vary freely.

Lemma (Degeneracy of the raw objective)

$\vartheta \mapsto \mathrm{CBO}_s(\eta, \vartheta)$ is strictly increasing on its admissible domain for both signs. The optimum sits only at the boundary ($\vartheta \to 0^+$ for $\overline{\mathrm{CBO}}$, $\vartheta \to \infty$ for $\underline{\mathrm{CBO}}$).

Idea

Proof sketch: by Danskin's theorem, with inner extremiser $\theta_s := \mathrm{argext}_s\{-\hat{\ell}(\theta) + \frac{\vartheta}{2}\|\theta - \eta\|^2\}$ and Newton step $u_s := -H_s^{-1}\nabla\hat{\ell}(\eta)$ ($\theta_s - \eta \approx s\, u_s$): $\partial_\vartheta \mathrm{CBO}_s(\eta, \vartheta) = \frac{1}{2}\|u_s\|^2 > 0$ ($u_s \neq 0$ away from stationarity). No interior stationary point can exist.

Exactly the collapse observed when fitting the Gaussian candidate in the landscape analysis: the raw CBO alone cannot identify an interior precision.

A Geometric Penalty via Level-Set Log-Volume

Definition (Level-set volume)

For $g_{\eta,S} = \mathcal{N}^*(\theta; \eta, S^{-1})$, the level set $\{g_{\eta,S} \geq \alpha\}$ is an ellipsoid of volume

$$\mathrm{vol}_\alpha(S) = V_d(-2\log\alpha)^{d/2}(\det S)^{-1/2} \implies \log \mathrm{vol}_\alpha(S) = -\tfrac{1}{2}\log\det S + c(\alpha, d)$$

- **No normaliser in possibility theory**: $\sup g = 1$, no natural entropy penalises concentration
- **Geometric cost**: penalise by the log-volume of the candidate's level sets *(suggested by S. Wongso)*, a natural proxy for informativeness, optimisation direction depending on which bound is being enforced
- **Invariance**: $c(\alpha, d)$ does not depend on S, choice of $\alpha \in (0, 1)$ irrelevant to optimization

The Penalty Selects an Extremal-Volume Candidate

Theorem (Extremal-volume selection)

At multiplier $\mu = 1$ (up to additive constants), the Lagrangian of:

(a) $\min_{\eta,S} \log \mathrm{vol}_\alpha(S)$ *s.t.* $\overline{\mathrm{CBO}}(g_{\eta,S}) \leq \log Z_{\max}$ *($g \geq g^\star$) is* PCBO_s *($s = +1$).*

(b) *For a target level $c \leq \log Z_{\max}$:* $\max_{\eta,S} \log \mathrm{vol}_\alpha(S)$ *s.t.* $\underline{\mathrm{CBO}}(g_{\eta,S}) \geq c$ *($g \leq g^\star$ at $c = \log Z_{\max}$) is* PCBO_s *($s = -1$).*

Idea

Proof sketch: form the Lagrangian $L = \log \mathrm{vol}_\alpha(S) + \mu(\mathrm{CBO} - \mathrm{cst})$, inject $\log \mathrm{vol}_\alpha(S) = -\frac{1}{2} \log\det S + c$, evaluate at $\mu = 1$.

Geometric meaning

Prevents trivial solutions: $g \geq g^\star$ trivial for 1_Θ ($\mathrm{vol} = \infty$), $g \leq g^\star$ trivial for a Dirac spike ($\mathrm{vol} = 0$). The penalty isolates the most informative non-trivial candidate.

The Penalty Repairs the Degeneracy

Proposition (Penalty breaks monotonicity)

For any $\zeta > 0$ (penalty coefficient, generalising $\mu = 1$), $\mathrm{PCBO}_s(\eta, \cdot)$ has an interior stationary point: a unique global minimum for $s = +1$ ($\overline{\mathrm{CBO}}$), a unique global maximum for $s = -1$ ($\underline{\mathrm{CBO}}$), in the exact-quadratic case. Unlike the raw objective, $\partial_\vartheta \mathrm{PCBO}_s$ now diverges at both ends of the admissible domain.

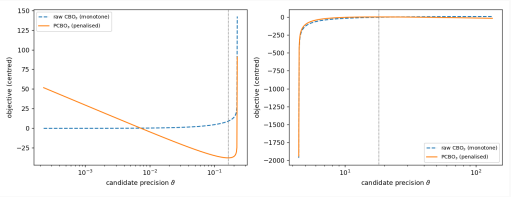


Figure. Raw versus penalised objective on the isotropic toy problem, far regime.

Four precision structures: isotropic ($S = \vartheta I$), diagonal ($S = \mathrm{diag}(\boldsymbol{\vartheta})$), low-rank-plus-diagonal (LRD), block-diagonal (BD).

Structure	Penalty $-\frac{1}{2} \log \det S$
Isotropic	$-\frac{d}{2} \log \vartheta$
Diagonal	$-\frac{1}{2} \sum_{j=1}^{d} \log \vartheta_j$
LRD	$\frac{1}{2}(d-1) \log \psi + \frac{1}{2} \log(\psi + \lVert u_s^{\mathrm{LRD}} \rVert^2)$
BD	$\sum_\ell \left(\frac{1}{2}(d_\ell - 1) \log \psi_\ell + \frac{1}{2} \log(\psi_\ell + \lVert u_{s,\ell}^{\mathrm{BD}} \rVert^2) \right)$

Table. Specificity penalty for each precision structure.

Isotropic and Diagonal Stationarity

Isotropic precision $\vartheta \in \mathbb{R}_{>0}$, diagonal precision $\boldsymbol{\vartheta} \in \mathbb{R}^d_{>0}$. Setting $\partial_\vartheta \mathrm{PCBO}_s = 0$ (isotropic) or $\partial_{\vartheta_j} \mathrm{PCBO}_s = 0$ (diagonal) gives the stationarity condition:

Proposition (Optimal precision from the Newton step)

$$\vartheta^\star \approx \frac{d}{\|u_s\|^2}, \qquad \vartheta_j^\star \approx \frac{1}{(u_s^{\mathrm{diag}})_j^2}, \qquad u_s := -H_s^{-1} \nabla \hat{\ell}(\eta)$$

This condition is **implicit** (H_s depends on ϑ), solved by damped Picard iteration (Krasnoselskii–Mann relaxation [8]), written in isotropic (scalar) form. The diagonal case applies the same map coordinatewise:

Damped Picard map, with anti-collapse/positivity cap

$$\vartheta_{t+1} = T_w(\vartheta_t) := (1-w)\vartheta_t + w\, \mathcal{C}_s\big(\Phi(\vartheta_t)\big), \qquad \Phi(\vartheta) := \frac{d}{\|u_s(\vartheta)\|^2}$$

$$\mathcal{C}_{+1}(\vartheta) := \max\big(\min(\vartheta,\ m/r,\ m(1-\epsilon_{\mathrm{rel}})),\ v_{\min}\big), \quad \mathcal{C}_{-1}(\vartheta) := \max\big(\min(\vartheta,\ M/r),\ M(1+\epsilon_{\mathrm{rel}})\big)$$

$m = \min_j(h_j + \delta)$, $M = \max_j(h_j + \delta)$: curvature bounds. r: anti-collapse divisor. ϵ_{rel}: positivity margin. $v_{\min}$: floor.

$\mathcal{C}_s$ enforces $H_s \succ 0$ and prevents Dirac collapse, keeping ϑ inside an invariant interval, a known device in natural-gradient VI [9].

Linear Convergence of the Damped Picard Map

Same scalar form as before, extended coordinatewise to the diagonal case. The caps confine the iterates to $\mathcal{I}_{+1} \coloneqq \left[v_{\min},\ m(1-\epsilon_{\mathrm{rel}})\right]$ and $\mathcal{I}_{-1} \coloneqq \left[M(1+\epsilon_{\mathrm{rel}}),\ M/r\right]$.

Theorem (Picard convergence)

Let $|\Phi'(\vartheta)| \le L_\Phi$ on an invariant interval $\mathcal{I}_s$, $\kappa(w) \coloneqq 1 + w(L_\Phi - 1)$. If $L_\Phi < 1$, T_w is a $\kappa(w)$-contraction for every $w \in (0,1]$. By Banach's fixed-point theorem it admits a unique fixed point $\vartheta^\dagger \in \mathcal{I}_s$, with $|\vartheta_t - \vartheta^\dagger| \le \kappa(w)^t |\vartheta_0 - \vartheta^\dagger|$.

Idea

Proof sketch: the cap $\mathcal{C}_s$ is a projection onto $\mathcal{I}_s$, hence non-expansive. The mean value theorem bounds Φ's Lipschitz constant by L_Φ on $\mathcal{I}_s$.

Quantity evaluated in the experiments

ϑ_0 initialised symmetrically close to the boundary, K damped Picard steps per mode update η_t. In practice, L_Φ estimated by the explicit bound $L_\Phi = 2d\|g\|^2/(u_{\min}(\eta)^4 a_{\min}^3)$, $g \coloneqq \nabla\hat{\ell}(\eta)$, $a_{\min} \coloneqq \epsilon_{\mathrm{rel}} \min_j(h_j + \delta)$, $u_{\min}(\eta) \coloneqq \min_{\mathcal{I}_s} \|u_s\| > 0$, degrading near stationary points.

Full and Block-Diagonal Precisions

Beyond isotropic and diagonal: a general precision $S \in \mathcal{S}^d_{++}$, block-diagonal $S = \bigoplus_\ell S_\ell$.

Proposition (Rank-one singularity)

Exact stationarity gives $S_\star^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top$, rank one and singular for $d > 1$. Regularise to a low-rank-plus-diagonal (LRD) form, similar to low-rank covariance approximations in VI [10]:

$$S^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top + \psi I, \qquad \psi > 0$$

ψ must certify $H_s \succ 0$ [9], signs behaving differently (BD applies the same per block, with $m_\ell, M_\ell, u_{s,\ell}^{\mathrm{BD}}$):

$s = +1$, unconditional

$\lambda_{\max}(S) < m \iff \psi > 1/m$: set $\psi = 1/m + a$ ($a > 0$: margin above $1/m$). Sufficient for any curvature.

$s = -1$, conditional

Feasible only if $\|u_s^{\mathrm{LRD}}\|^2 < 1/M$: set $\psi = \frac{1}{\varphi}\left(\frac{1}{M} - \|u_s^{\mathrm{LRD}}\|^2\right)$ ($\varphi > 1$: multiplicative damping factor). Fails early in training.

When infeasible (LRD: globally, BD: per block), flip to $s_{\mathrm{e}} = +1$ until $\|u_s^{\mathrm{LRD}}\|^2$ drops below $(1 - \xi)/M$ ($\xi \in (0, 1)$: margin avoiding a near-zero ψ), then switch back to $s_{\mathrm{e}} = -1$: the standard warm-up principle of starting broad before tightening toward the target [11].

A Chain of Results

Common to all four structures

Raw CBO degenerate $\rightarrow$ log-volume penalty selects extremal-volume candidate

$\downarrow$

Isotropic / Diagonal

Implicit fixed point $\rightarrow$ damped Picard step under caps (linear convergence on $\mathcal{I}_s$) $\rightarrow$ `CBOpt-iso` / `CBOpt-diag`

LRD / BD

Stationarity singular $\rightarrow$ LRD regularisation $\rightarrow$ sign-asymmetric ψ safeguards (unconditional for $s = +1$, warm-up flip for $s = -1$) $\rightarrow$ `CBOpt-LRD` / `CBOpt-BD`

SAM Application, Algorithms, and Experimental Validation

Lower CBO and Local Robustness: Connecting to SAM

An application of the framework above, first derived by Shelvia Wongso [12], connecting to a classical robust-optimisation setting.

Sharpness-Aware Minimization (SAM) [13] minimizes local worst-case loss (here, $\delta = 0$):

$$\mathcal{E}_{\mathrm{SAM}}(\eta;\rho) \coloneqq \sup_{\|\varepsilon\|_2 \le \rho} \ell(\eta+\varepsilon)$$

Proposition (Indicator candidate is SAM, 12)

For $\Theta \subseteq \mathbb{R}^d$ closed, ℓ continuous, flat prior $\pi \equiv 1$, and $g_{\eta,\rho}(\theta) \coloneqq 1[\|\theta-\eta\|_2 \le \rho]$: $-\log g_{\eta,\rho} = +\infty$ outside the ball forces the $\underline{\mathrm{CBO}}$ ($s=-1$) infimum to match worst-case loss.

$$\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\sup_{\|\theta-\eta\|_2 \le \rho} \ell(\theta) \implies \arg\max_{\eta} \underline{\mathrm{CBO}}(g_{\eta,\rho}) = \arg\min_{\eta} \mathcal{E}_{\mathrm{SAM}}(\eta;\rho)$$

If $\nabla\ell(\eta) \neq 0$ and $B(\eta,\rho) \subseteq \Theta$, the first-order step is $\varepsilon^\star_{\mathrm{SAM}} = \rho\,\nabla\ell(\eta)/\|\nabla\ell(\eta)\|_2$.

Exact identity of objectives, not just stationary points. Unlike the probabilistic SAM–Bayes link, which only holds as a convex relaxation [14].

Gaussian Relaxation and the Adaptive Radius

Limitation: ρ is fixed manually in SAM, motivating a Gaussian candidate with a learned radius.

Proposition (Gaussian relaxation of SAM, 12)

For $g_{\eta,\vartheta} = \mathcal{N}^*(\cdot; \eta, \vartheta^{-1} I)$, the lower CBO relaxes the hard ball into a quadratic penalty:

$$\underline{\mathrm{CBO}}(g_{\eta,\vartheta}) = -\sup_{\theta \in \mathbb{R}^d} \left\{ \ell(\theta) - \tfrac{\vartheta}{2} \|\theta - \eta\|_2^2 \right\}$$

For $L < \vartheta$ (L: Lipschitz constant of $\nabla \ell$), the exact maximiser $u^\star$ exists and is unique. Its first-order surrogate is $u^\star_{\mathrm{soft}} = \vartheta^{-1} \nabla \ell(\eta)$, exact when ℓ is affine near η.

Superlevel-set bridge: $\{g_{\eta,\vartheta} \geq \alpha\} = \bar{B}(\eta, \rho_\alpha)$, $\rho_\alpha := \vartheta^{-1/2} \sqrt{-2 \log \alpha}$, giving $\rho_\alpha = \vartheta^{-1/2}$ at $\alpha = e^{-1/2}$.

Adaptive SAM radius derived from the learned precision

With $\vartheta^\star = d / \|u_{-1}\|_2^2$ and $u_{-1} \approx -H_{-1}^{-1} \nabla \hat{\ell}(\eta)$:

$$\rho^\star(\eta) \approx \frac{1}{\sqrt{d}} \left\| H_{-1}^{-1} \nabla \hat{\ell}(\eta) \right\|_2$$

- **Direction vs norm**: $u^\star_{\mathrm{soft}}$ shares SAM's direction, but its norm scales as $\vartheta^{-1} \|\nabla \ell(\eta)\|$, vs SAM's fixed ρ
- **Hyperparameter trade-off**: replaces direct tuning of ρ with the spread safeguards $(r, \epsilon_{\mathrm{rel}})$

Algorithms and Complexity

`uCBOpt`/`lCBOpt` $\times$ {iso, diag, LRD, BD} ($s = \pm 1$) differ in storage and inversion cost:

Structure	Storage	$H_s^{-1}(\bar{g}_t + \delta\eta_t)$
Isotropic	$O(1)$	$O(d)$
Diagonal	$O(d)$	$O(d)$
LRD	$O(d)$	$O(d)$ (Sherman–Morrison)
Block-diagonal	$O(d)$	$O(d)$

Table. Per-mode-step storage and inversion cost of each precision structure.

- **Predicted order**: $\mathrm{LRD} \approx \mathrm{BD} < \mathrm{iso} \approx \mathrm{diag}$, the closed-form pair ($\psi$ in closed form, no iteration) against the K-iteration pair ($O(Kd)$)
- **LRD inversion**: $S^{-1} = uu^\top + \psi I$ stored as $(u, \psi) \in \mathbb{R}^d \times \mathbb{R}_{>0}$, never as a dense $d \times d$ matrix. Sherman–Morrison solves $H_s^{-1} z$ in $O(d)$, avoiding the $O(d^3)$ cost of a naive inversion

Toy Problem, Sanity Checks and Genuine Findings

Exact quadratic loss, $d = 30$, closed form, no minibatch or EMA noise.

Sanity checks (hold by construction)

- Regularity (assumption in the report): $\theta_s(\vartheta)$ finite and unique across the admissible grid, both signs
- Monotonicity and its repair: raw CBO strictly monotone, penalty gives an interior stationary point (already shown)
- SAM direction: $\varepsilon^\star_{\mathrm{SAM}}$ and $u^\star_{\mathrm{soft}}$ collinear to machine precision ($\cos = 1.000000$)

Genuine findings

- Penalty coefficient sensitivity: $\vartheta^\star$ moves 2.8× ($s = +1$, sub-linear) and 6.4× ($s = -1$, more sensitive) over a 16× coefficient range, justifying $\zeta = d/2$ rather than a hand-picked value
- Dimension sweep ($d \in [3.5 \times 10^3, 1.9 \times 10^5]$): confirms the predicted pairing, refined to $\mathrm{LRD} < \mathrm{BD} < \mathrm{iso} < \mathrm{diag}$ and measured factor $\sim 2\times$ rather than K=5×, likely GPU overhead

Reproduction and Benchmark on LeNet-5 and Fashion-MNIST

LeNet-5 ($d = 44{,}426$), Fashion-MNIST (in-domain) vs EMNIST-Letters (OOD), 100 epochs, 3 seeds.

Optimiser	Acc. (%) ↑	NLL ↓	ECE ↓	FPR@95 ↓	AUROC ↑
`uCBOpt-iso` (learned ϑ)	91.2 ± 0.1	0.257 ± 0.002	0.022 ± 0.001	$\mathbf{0.263 \pm 0.015}$	$\mathbf{0.793 \pm 0.018}$
`uCBOpt-diag` (learned $\boldsymbol{\vartheta}$)	$\mathbf{91.5 \pm 0.1}$	$\mathbf{0.252 \pm 0.001}$	0.020 ± 0.002	0.292 ± 0.018	0.758 ± 0.018
`uCBOpt-BD` (learned S)†	90.8 ± 0.2	0.256 ± 0.001	$\mathbf{0.014 \pm 0.001}$	0.289 ± 0.017	0.755 ± 0.018
fixed `uCBOpt` [1]	90.9 ± 0.2	0.272 ± 0.004	0.025 ± 0.001	0.263 ± 0.005	0.795 ± 0.005
IVON (64 MC samples) [15]	$\mathbf{91.0 \pm 0.2}$	$\mathbf{0.258 \pm 0.003}$	$\mathbf{0.019 \pm 0.001}$	$\mathbf{0.241 \pm 0.009}$	$\mathbf{0.812 \pm 0.012}$
AdamW [16]	89.6 ± 0.3	0.296 ± 0.005	0.022 ± 0.006	0.335 ± 0.005	0.722 ± 0.016

Table. Reproduction and learned-precision benchmark. All `CBOpt` variants use a single deterministic pass.
† *Seeds 0, 1 (seed 2 diverged at epoch 16).*

- **Reproduction**: all reference baselines matched within 2σ of Singh et al. [1]
- **In-domain gains**: learned precision improves Acc, NLL, and calibration, from a **single pass** (64× cheaper)
- **OOD trade-off**: the three extensions score below fixed `uCBOpt` and IVON on FPR@95/AUROC. Wider seed variance on OOD affects every precision-learning method, IVON included, matching the underspecification pattern [17]

Geometry and Safeguards

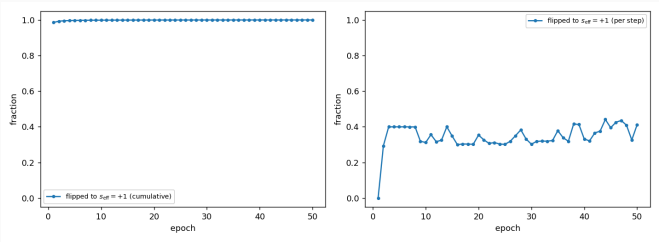


Figure. Fraction of steps or blocks flipped to $s_{\text{eff}} = +1$ for `lCBOpt-LRD` and `lCBOpt-BD`.

- $s = +1$, **iso/diag**: $H_s \succ 0$ before capping held on 0.0% of steps. The cap is the mechanism, capping ϑ to $(1 - \epsilon_{\text{rel}})(h_j + \delta)$, the same form as `uCBOpt-adapt`'s hand-designed γc_t envelope [1], matching it numerically
- $s = -1$, **LRD**: the global feasibility condition ($\|u\|^2 < 1/M$ across all d coordinates at once) never holds at network scale. `lCBOpt-LRD` flips on 100% of steps
- $s = -1$, **BD**: per-block feasibility is far easier to satisfy, only 34.5% of blocks flip. 65.5% still pursue the $s = -1$ objective, the repair block-diagonalisation was designed to provide

Picard Contraction at Network Scale

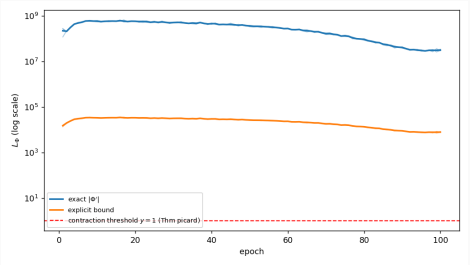


Figure. $|\Phi'|$ *and* L_Φ, `uCBOpt-iso` ($s = +1$).

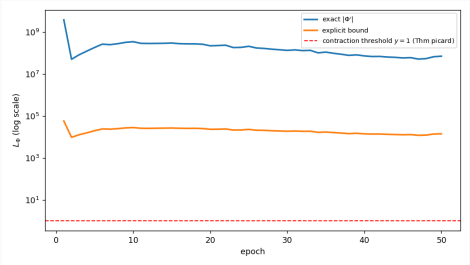


Figure. $|\Phi'|$ *and* L_Φ, `lCBOpt-iso` ($s = -1$).

- **Magnitude**: $|\Phi'|$ and L_Φ both exceed 1 from epoch 1, reaching 3×10^7 ($s = +1$) and 7×10^7 ($s = -1$) by the end of training at $d = 44{,}426$. $L_\Phi = 2d\|g\|^2/(u_{\min}(\eta)^4 a_{\min}^3)$, $g := \nabla\hat{\ell}(\eta)$, $a_{\min} := \epsilon_{\mathrm{rel}} \min_j(h_j + \delta)$, $u_{\min}(\eta) := \min_{\mathcal{I}_s} \|u_s\| > 0$
- **Caveat**: the convergence theorem's sufficient condition fails. But ϑ sits at the cap's boundary rather than converging to an interior fixed point, so it never tests the regime the theorem certifies

SAM Application and Sensitivity Results

The identity $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\mathcal{E}_{\mathrm{SAM}}(\eta;\rho,0)$ is exact, checked on a trained checkpoint. The gap grows as $O(\rho^2)$, as expected from the first-order approximation $\varepsilon^{\star}_{\mathrm{SAM}}$: 1.79×10^{-3} at $\rho = 0.02$, up to 1.12×10^{-1} at $\rho = 0.1$.

$\pm 5\%$ sweep of r ($r_{\mathrm{nom}} = 0.50$) and ρ ($\bar{\rho}_{\mathrm{nom}} = 2.2165$, learned radius's mean), one seed, 50 epochs.

	$\times 0.95$	$\times 1.0$	$\times 1.05$
`lCBOpt-iso` (r)	0.8865	0.8876	0.8904
standard SAM [13] (ρ)	0.1373	0.2168	0.2399

Table. Accuracy under a $\pm 5\%$ sweep, learned versus standard SAM radius.

- **Sensitivity**: $\Delta_{\mathrm{adapt}} = 0.0039$ vs $\Delta_{\mathrm{SAM}} = 0.1026$, $26\times$ higher sensitivity for standard SAM
- **Radius scale**: $\bar{\rho}_{\mathrm{nom}} \approx 2.2$ is an order of magnitude above the SAM literature's typical $\rho \in [0.01, 0.2]$ [13]
- **Caveat**: ϑ saturates the anti-collapse bound rather than the interior optimum, so both are compared in suboptimal regimes

Discussion and Conclusion

Four Operational Limitations

1. **Safeguard infeasible**: the $s = -1$ safeguard fails on 100% of steps for `lCBOpt-LRD`. `lCBOpt-BD` partially repairs this (65.5% of blocks), but `uCBOpt-BD` remains numerically fragile (one benchmark seed diverged)
2. **Precision stays at the boundary**: `uCBOpt-iso` and `uCBOpt-diag` never reach the interior fixed point, sitting at the cap throughout training

   $\downarrow$
3. **Picard contraction fails**: $L_\Phi \sim 10^7$ to 10^8 at network scale for `uCBOpt-iso` and `uCBOpt-diag`
4. **SAM radius too large**: $\bar{\rho} \approx 2.2$ from `lCBOpt-iso` vs $\rho \in [0.01, 0.2]$ in the literature [13]

Conclusion and Open Questions

Contributions

1. Reproduced Singh et al. [1] within two standard deviations
2. Diagnosed misspecification, not divergence, as the source of candidate-target gaps
3. Learned the precision via a log-volume penalty: damped Picard fixed points for iso/diag, regularised closed-form safeguards for LRD/BD
4. Connected the lower CBO to SAM [13] and derived an adaptive perturbation radius
5. Assembled eight practical optimisers with a complexity discussion

Open questions

Multimodal/non-parametric candidates, annealing p, volume-penalty guarantees beyond the quadratic case, optimality of the LRD regulariser, joint (η, ϑ) convergence [18], whether the Gaussian relaxation for the learned SAM radius is theoretically sound.

Thank you

Questions?

Detail: The Two Debates

Framework: epistemic uncertainty on a fixed $\theta^\star \in \Theta$, no repeatable draw. Both examples below are illustrations from the literature, not claims that probability is wrong in general.

Debate 1: improper priors

On $\Theta = (0, \infty)$ with $\pi(\theta) \propto 1$: with $X \mid \theta \sim \mathrm{Unif}(0, \theta)$, the posterior is $\pi(\theta \mid x) \propto \theta^{-1} 1_{\theta \geq x}$, whose normalising integral $\int_x^\infty \theta^{-1} \, \mathrm{d}\theta$ diverges: an improper posterior from an improper prior. Jeffreys' prior $\pi_J \propto \sqrt{\det I(\theta)}$ is invariant under reparametrisation, but for $X \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$, $I(\mu) = 1/\sigma^2$ gives $\pi_J(\mu) \propto 1$, still improper on $\mathbb{R}$. Since I depends on the likelihood, the representation of ignorance ends up depending on the data-generating model.

Debate 2: probability dilution (false confidence)

For $A \subset \Theta$ bounded and $\theta^\star \sim \mathcal{N}(\mu, \sigma^2)$: as $\sigma \to \infty$, $\mathbb{P}(\theta^\star \in A) \to 0$ even when $\mu \in A$ always. Balch, Martin, Ferson [2] call this false confidence: for some false $A \not\ni \theta^\star$, probability can assign arbitrarily high belief with arbitrarily high probability under the true model. Not every Bayesian agrees this is a flaw: a posterior probability was never meant to certify this kind of coverage, a frequentist-style guarantee.

Duality: Why Two Objectives, Not One

Framework: the maxitive Donsker–Varadhan formula is a min-max equality, $\sup_g \inf_\theta\{\cdots\} = \inf_g \sup_\theta\{\cdots\} = \log Z_{\max}$, read from each side separately, unlike the classical formula's single supremum.

Why the duality appears

Each side of the equality gives its own variational characterisation: maximising the inner infimum gives $\underline{\mathrm{CBO}}$, minimising the inner supremum gives $\overline{\mathrm{CBO}}$. Nothing forces these two readings to coincide pointwise in g away from the saddle, so two objectives are needed rather than one. The classical formula has no such second reading: $\mathrm{KL}(q\|p)$ is a single divergence from a fixed target p, stated as one supremum over ρ, with no infimum-based formula for the same quantity.

Why $g^\star_{\max}$ is still the right answer

Zellner's theorem [19] shows Bayes' rule is the unique information-processing rule that neither creates nor destroys information, so the probabilistic formula recovers the Bayesian posterior because it is optimal by this criterion. A possibilistic analogue of this question [20], replacing Zellner's scalar information measure by a pointwise score compared through the specificity order $\preceq$, identifies $g^\star_{\max}$ as what that criterion selects, the maxitive formula is the variational expression of this same optimality result.

Proof: The Sandwich Identity

Framework: the exact identity tying $\underline{\mathrm{CBO}}$ and $\overline{\mathrm{CBO}}$ to $\log Z_{\max}$ via the max-relative entropy $D_{\max}$.

Theorem

$$\log Z_{\max} = \underline{\mathrm{CBO}}(g) + D_{\max}(g \| g^\star_{\max}) = \overline{\mathrm{CBO}}(g) - D_{\max}(g^\star_{\max} \| g)$$

Proof. By definition, $\underline{\mathrm{CBO}}(g) = \inf_\theta\{-\ell(\theta) - \log \frac{g(\theta)}{\pi(\theta)}\} = \inf_\theta\{-\hat{\ell}(\theta) - \log g(\theta)\}$, using $\hat{\ell} = \ell - \log \pi$. Writing $g^\star_{\max} = e^{-\hat{\ell}}/Z_{\max}$ and substituting $-\hat{\ell}(\theta) = \log g^\star_{\max}(\theta) + \log Z_{\max}$ inside the infimum gives $\underline{\mathrm{CBO}}(g) = \log Z_{\max} + \inf_\theta \log \frac{g^\star_{\max}(\theta)}{g(\theta)} = \log Z_{\max} - \sup_\theta \log \frac{g(\theta)}{g^\star_{\max}(\theta)} = \log Z_{\max} - D_{\max}(g \| g^\star_{\max})$, rearranging gives the first equality. The second follows symmetrically from the definition of $\overline{\mathrm{CBO}}$ as a supremum instead of an infimum, swapping the roles of g and $g^\star_{\max}$ inside $D_{\max}$.

Consequence

Maximising $\underline{\mathrm{CBO}}$ minimises $D_{\max}(g \| g^\star_{\max})$ (mode-seeking, under-estimation). Minimising $\overline{\mathrm{CBO}}$ minimises $D_{\max}(g^\star_{\max} \| g)$ (support-covering, over-estimation). Both $D_{\max}$ terms vanish exactly when $g = g^\star_{\max}$, recovering the sandwich $\underline{\mathrm{CBO}}(g) \le \log Z_{\max} \le \overline{\mathrm{CBO}}(g)$.

Reminder: Probabilistic vs Possibilistic VI

	Probabilistic VI	**Possibilistic VI**
Candidate space	$\mathcal{P}(\Theta)$ $(\int q = 1)$	$\mathcal{F}(\Theta)$ $(\sup g = 1)$
Uncertainty	aleatoric	epistemic
Non-informative prior	improper $(\int = \infty)$	1_Θ, proper
Log-evidence	$\log Z_{\mathrm{add}} = \log \int e^{-\hat{\ell}(\theta)}\,\mathrm{d}\theta$	$\log Z_{\max} = \log \sup e^{-\hat{\ell}(\theta)}$
Variational formula	$\log Z_{\mathrm{add}} = \sup_q \mathrm{ELBO}(q)$	$\log Z_{\max} = \sup_g \underline{\mathrm{CBO}}(g) = \inf_g \overline{\mathrm{CBO}}(g)$
Divergence	$\mathrm{KL}(q \Vert \pi(\cdot \mid x_{1:n}))$	$D_{\max}(g \Vert g^\star_{\max})$ and $D_{\max}(g^\star_{\max} \Vert g)$
$D = 0 \iff$	$q = \pi(\cdot \mid x_{1:n})$	$g \preceq g^\star_{\max}$ and $g^\star_{\max} \preceq g$
Unique optimiser?	yes	no, intervals $\{g \preceq g^\star_{\max}\}$ and $\{g^\star_{\max} \preceq g\}$
Expectation	mean $\int \theta q(\mathrm{d}\theta)$	mode $\arg\max_\theta g(\theta)$
Fisher information	$\mathbb{E}[\nabla \log p \, \nabla \log p^\top]$ (global)	$-\nabla^2 \log g\vert_{\mathsf{mode}}$ (local)

Table. Structural correspondence between probabilistic and possibilistic variational inference.

Derivation: The Signed Newton-Like Update

Framework: possibilistic exponential family $g_\lambda(\theta) = \exp(\lambda^\top T(\theta) - A(\lambda) - B(\theta))$, reparametrised by the mode $\eta = \nabla A^\dagger(\lambda)$, signed curvature $H_s(\eta) := s(\nabla^2 \hat{\ell}(\eta) - \nabla^2 A_0(\eta))$.

Derivation. By Danskin's theorem applied to the inner extremum defining CBO_s, $\nabla_\eta \mathrm{CBO}_s(\eta) = \nabla^2 A_0(\eta)(\eta - \theta_s)$, where θ_s is the inner extremiser. Replacing the inner objective by its second-order Taylor expansion around η and minimising the resulting surrogate of $s\,\mathrm{CBO}_s$ via natural gradient descent (preconditioning by the local curvature rather than the identity) yields the update valid when $H_s \succ 0$:

$$\eta_{t+1} = \eta_t - \rho_t\, H_s(\eta_t)^{-1} \nabla \hat{\ell}(\eta_t).$$

Isotropic Gaussian specialisation

With candidate precision ϑ and L^2 prior precision δ, the signed curvature becomes $H_s = s(\nabla^2 \ell + (\delta - \vartheta) I)$. Approximating $\nabla^2 \ell$ by an EMA of the squared stochastic gradient, $h_t \leftarrow \beta_2 h_{t-1} + (1 - \beta_2)\hat{g}_t \odot \hat{g}_t$, gives the implementable form $\eta_{t+1} = \eta_t - \rho_t\, s\, \dfrac{\bar{g}_t + \delta \eta_t}{h_t + \delta - \vartheta}$, coordinatewise, with $\bar{g}_t$ the bias-corrected EMA gradient.

Landscapes: Sandwich Decomposition Across Tractable Targets

Table. Sandwich decomposition across tractable targets. Systematic gap explosion under misspecification.

Case	Family	$\log Z_{\max}$	$\underline{\mathrm{CBO}}$	$\overline{\mathrm{CBO}}$	$D_{\max}(g\|g^\star)$	$D_{\max}(g^\star\|g)$
Gaussian (sanity check)	$\mathcal{N}^*$	-0.20	-0.76	0.30	0.567	0.498
Bern.–Beta	$\mathcal{N}^*$	-15.16	-152.49	-15.16	137.33	0.0026
	Beta^*	-15.16	-15.16	-15.16	3.1×10^{-5}	2.3×10^{-5}
Laplace	$\mathcal{N}^*$	-17.09	-17.44	632.99	0.346	650.08
	Lap^*	-17.09	-110.99	-16.68	93.89	0.418
Bimodal	$\mathcal{N}^*$ (μ=2)	-9.40	-43.43	6.40	34.03	15.80

- **Gaussian**: residuals come from grid spacing δ and domain truncation on $\mathbb{R}$
- **Bernoulli–Beta**: support mismatch ($g_{\mathcal{N}^*} > 0$ outside $(0,1)$)
- **Laplace**: tail mismatch (see figure)
- **Bimodal**: single mode cannot be both mode-seeking and support-covering

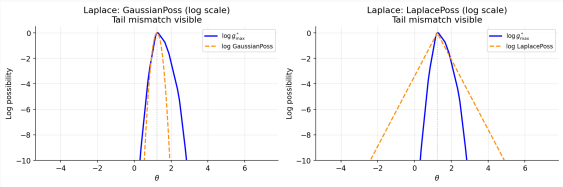


Figure. Laplace case: tail mismatch in log-scale ($\log g^\star_{\max}$ *decays faster than candidates).*

Landscapes: The Order-p Divergence Test

$$D_p(g\|f) \coloneqq \begin{cases} \dfrac{1}{p}\log \displaystyle\sum_{\theta\in\mathrm{supp}(f)} \exp\big(p(\log g(\theta) - \log f(\theta))\big) & \text{if } \mathrm{supp}(g) \subseteq \mathrm{supp}(f) \\ +\infty & \text{otherwise} \end{cases}$$

- **Convergence**: $D_p \xrightarrow[p\to\infty]{} D_{\max}$ (soft LogSumExp relaxation).
- **Bounds**: $D_{\max} \le D_p \le D_{\max} + \frac{1}{p}\log N$: mismatched gaps stay constant (dominated by one point).
- **Conclusion**: comparative ranking is invariant, gaps stem from family misspecification, not $D_{\max}$.
- **Left unexplained**: directly maximising CBO over precision collapses ($\vartheta \to \infty$).

Table. Order-p divergence at $p = 1$ (softest) and $p = 50$, against the $D_{\max}$ reference.

Case	Direction	$p = 1$	$p = 50$	$D_{\max}$ (ref)
Gaussian	$D_p(g\|g^\star)$	6.29	0.61	0.57
	$D_p(g^\star\|g)$	6.23	0.54	0.50
Bern.–Beta	$D_p(g\|g^\star)$	137.33	137.33	137.33
	$D_p(g^\star\|g)$	5.64	0.10	0.003
Laplace	$D_p(g\|g^\star)$	3.44	0.36	0.35
	$D_p(g^\star\|g)$	650.09	650.08	650.08
Bimodal	$D_p(g\|g^\star)$	35.00	34.03	34.03
	$D_p(g^\star\|g)$	19.57	15.84	15.80

Proof: The Log-Sum-Exp Bound on D_p

Framework: the order-p softening $D_p(g\|f) := \frac{1}{p}\log\sum_{\theta\in\mathrm{supp}(f)}\exp(p(\log g(\theta) - \log f(\theta)))$ on an N-point grid, compared to $D_{\max}(g\|f) := \sup_\theta \log\frac{g(\theta)}{f(\theta)}$.

Proposition

$D_{\max}(g\|f) \le D_p(g\|f) \le D_{\max}(g\|f) + \frac{1}{p}\log N$.

Proof. Let $M := \max_\theta\{\log g(\theta) - \log f(\theta)\} = D_{\max}(g\|f)$. Lower bound: the sum over the grid is at least its largest term, $\sum_\theta \exp(p(\log g(\theta) - \log f(\theta))) \ge \exp(pM)$, so $D_p \ge \frac{1}{p}\log\exp(pM) = M = D_{\max}$. Upper bound: each of the N terms is at most $\exp(pM)$, so the sum is at most $N\exp(pM)$, giving $D_p \le \frac{1}{p}\log(N\exp(pM)) = M + \frac{1}{p}\log N = D_{\max} + \frac{1}{p}\log N$.

Consequence used in the report

As $p \to \infty$, the upper bound's slack $\frac{1}{p}\log N \to 0$, so $D_p \to D_{\max}$: a single dominant point makes D_p tight even at small p, while several comparable points inflate D_1 well above $D_{\max}$ without changing the eventual limit.

Candidate Fitting: Protocol and Analytical Expressions

Fitting protocol ($N = 500$ points, to avoid boundary collapse): location/scale by grid maximiser and finite difference (Gaussian, Laplace), OLS on the log-linear form (Beta).

Candidate possibility functions:

- Gaussian: $\mathcal{N}^*(\theta; \mu, \sigma^2) = \exp\left(-\frac{1}{2\sigma^2}(\theta - \mu)^2\right)$
- Laplace: $\mathrm{Lap}^*(\theta; \mu, b) = \exp\left(-|\theta - \mu|/b\right)$
- Beta: $\mathrm{Beta}^*(\theta; \alpha, \beta) = \dfrac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{\theta_m^{\alpha-1}(1-\theta_m)^{\beta-1}}$, $\theta_m = \dfrac{\alpha - 1}{\alpha + \beta - 2}$ (mode, valid for $\alpha, \beta > 1$)

Exact possibilistic posteriors $g^\star_{\max}(\theta)$:

- Gaussian: $g^\star_{\max}(\theta) = \exp\left(-\frac{1}{2\sigma^2_{\mathrm{post}}}(\theta - \mu_{\mathrm{post}})^2\right)$
- Bern.–Beta: $g^\star_{\max}(\theta) = \dfrac{\theta^{\alpha^\star-1}(1-\theta)^{\beta^\star-1}}{\theta_\star^{\alpha^\star-1}(1-\theta_\star)^{\beta^\star-1}}$, $\theta_\star = \dfrac{\alpha^\star - 1}{\alpha^\star + \beta^\star - 2}$ ($\alpha^\star, \beta^\star > 1$ here)
- Laplace: $g^\star_{\max}(\theta) = \exp\left(-\sum_{i=1}^n |x_i - \theta| + \sum_{i=1}^n |x_i - \hat{\theta}_{\mathrm{med}}|\right)$
- Bimodal: no closed form; $\ell(\theta)$ evaluated pointwise via log-sum-exp

Candidate Fitting: Ground-Truth Comparison

Table. Prior, ground truth, fitted parameters, and grid domain/resolution, per case.

Case	Prior	Ground truth	Fitted parameters	Θ range	δ
Gaussian (sanity)	$\mathcal{N}^*(\cdot; 0, 4)$	$\mu_{\text{post}} = 1.2352$, $\sigma_{\text{post}} = 0.1818$	$\hat{\mu} = 1.2455$, $\hat{\sigma} = 0.1818$	$[-10.0, 11.5]$	0.0431
Bern.–Beta	$\text{Beta}^*(2, 2)$	$\alpha^\star = 13$, $\beta^\star = 11$	$\hat{\alpha} = 13.000$, $\hat{\beta} = 11.000$	$(10^{-6}, 1-10^{-6})$	0.0020
Laplace	1_Θ	$\hat{\theta}_{\text{med}} = 1.2374$, $b^\star = 1.0$	$\hat{\mu} = 1.2374$, $\hat{b} = 0.4042$	$[-4.77, 7.23]$	0.0240
Bimodal	1_Θ	modes at ± 2, valley at 0	$\hat{\mu} = -2.13$, $\hat{\sigma} = 0.32$	$[-5.0, 5.0]$	0.0200

Key take-away: three finite-implementation effects recur throughout (harmless sampling noise, grid resolution, and domain truncation on unbounded Θ). When geometries match, only these finite-implementation effects remain, and discrepancies explode strictly under structural misspecification.

Theoretical Framework: Standing Assumptions

Stated for the isotropic case. Extended coordinatewise to $\boldsymbol{\vartheta} \in \mathbb{R}^d_{>0}$ for the diagonal case, and to the matrix $S \in \mathcal{S}^d_{++}$ for LRD and BD.

Assumption (Local regularity, curvature, and inner extremisers)

Let $\hat{\ell} = \ell - \log \pi$ with an L^2 prior $-\log \pi(\theta) = \frac{\delta}{2}\|\theta\|^2$.

(a) ***Smoothness**: ℓ is C^2 on a neighbourhood U of η (a.e. for ReLU networks [7]).*

(b) ***Curvature sign**: $H_s(\theta, \vartheta) = s(\nabla^2 \hat{\ell}(\theta) - \vartheta I) \succ 0$ on $U \times \mathcal{I}_s$.*

(c) ***Mode regularity**: at fixed ϑ, the extremiser $\theta_s(\eta)$ exists, is unique, lies in U, continuous in η.*

(d) ***Precision regularity**: at fixed η, $\theta_s(\vartheta)$ exists, is unique, lies in U with $\theta_s(\vartheta) \neq \eta$ ($u_s(\vartheta) \neq 0$), continuous in ϑ.*

Role of (d)

(a)–(c) are implicit in Singh et al. [1]. (d) is our addition: it serves two purposes, keeping $u_s(\vartheta) \neq 0$ for the stationarity condition $\vartheta^\star = d/\|u_s\|^2$ and the Picard map, and making Danskin's theorem applicable along all of $\mathcal{I}_s$, from which the penalty's monotonicity-breaking follows.

Proof: Degeneracy of the Raw Objective

Framework: isotropic candidate $g_{\eta,\vartheta}$, L^2 prior of precision δ, ϑ free. Valid under Assumption (d): $\theta_s(\vartheta)$ unique, interior, $\neq \eta$, continuous in ϑ on $\mathcal{I}_s$.

Lemma

$\vartheta \mapsto \mathrm{CBO}_s(\eta, \vartheta)$ is strictly increasing on its admissible domain for both signs. The optimum sits only at the boundary, $\vartheta \to 0^+$ for $\overline{\mathrm{CBO}}$ and $\vartheta \to \infty$ for $\underline{\mathrm{CBO}}$.

Proof. Under Assumption (d), for any admissible ϑ, $\theta_s(\vartheta)$ is unique, interior to U, and satisfies $\theta_s(\vartheta) \neq \eta$. By Danskin's envelope theorem applied to the inner supremum ($s = +1$) and infimum ($s = -1$), $\vartheta \mapsto \mathrm{CBO}_s(\eta, \vartheta)$ is differentiable, with

$$\frac{\partial}{\partial \vartheta} \mathrm{CBO}_s(\eta, \vartheta) = \tfrac{1}{2} \|\theta_s(\vartheta) - \eta\|^2 > 0.$$

Strictly positive everywhere, this makes $\mathrm{CBO}_s(\eta, \cdot)$ strictly increasing for both $s = +1$ and $s = -1$: minimising $\overline{\mathrm{CBO}}$ ($s = +1$) pushes $\vartheta \to 0^+$, while maximising $\underline{\mathrm{CBO}}$ ($s = -1$) pushes $\vartheta \to \infty$, precluding any interior stationary point.

Derivation: Volume of a Gaussian Level Set

Framework: $g_{\eta,S} = \mathcal{N}^*(\theta; \eta, S^{-1})$ with $S \in \mathcal{S}^d_{++}$, level set $\{g_{\eta,S} \geq \alpha\}$ for $\alpha \in (0,1)$.

Derivation. Since $g_{\eta,S}(\theta) \geq \alpha \iff (\theta-\eta)^\top S(\theta-\eta) \leq c$ with $c := -2\log\alpha > 0$, the level set is the ellipsoid $\{x^\top Sx \leq c\}$ (centred at η). Diagonalise $S = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$. The rotation $y = Q^\top x$ preserves volume and gives $x^\top Sx = \sum_i \lambda_i y_i^2 \leq c$, i.e. an axis-aligned ellipsoid with semi-axes $a_i = \sqrt{c/\lambda_i}$. Its volume is $V_d \prod_i a_i = V_d\, c^{d/2} \big(\prod_i \lambda_i\big)^{-1/2} = V_d\, c^{d/2} (\det S)^{-1/2}$, using $\det S = \det \Lambda$ and V_d the volume of the unit ball in $\mathbb{R}^d$. Substituting $c = -2\log\alpha$:

$$\operatorname{vol}_\alpha(S) = V_d(-2\log\alpha)^{d/2}(\det S)^{-1/2}, \qquad V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}.$$

Log-volume

Taking logs isolates the only S-dependent term:

$$\log \operatorname{vol}_\alpha(S) = -\tfrac{1}{2}\log\det S + \underbrace{\log V_d + \tfrac{d}{2}\log(-2\log\alpha)}_{c(\alpha,d),\ \text{independent of } S}.$$

Since $c(\alpha, d)$ carries all the α-dependence and none of the S-dependence, the choice of level α never changes which candidate has smaller log-volume.

Proof: Extremal-Volume Selection

Framework: any $S \in \mathcal{S}_{++}^d$, a general precision matrix, holding regardless of any structural constraint later placed on S.

Theorem

At multiplier $\mu = 1$ (up to additive constants), the Lagrangian of $\min_{\eta,S} \log \mathrm{vol}_\alpha(S)$ s.t. $\overline{\mathrm{CBO}}(g_{\eta,S}) \le \log Z_{\max}$ is PCBO_s ($s = +1$). For a target level $c \le \log Z_{\max}$, the Lagrangian of $\max_{\eta,S} \log \mathrm{vol}_\alpha(S)$ s.t. $\underline{\mathrm{CBO}}(g_{\eta,S}) \ge c$ is PCBO_s ($s = -1$).

Proof (a). The Lagrangian is $L(\eta, S, \mu) = \log \mathrm{vol}_\alpha(S) + \mu(\overline{\mathrm{CBO}}(g_{\eta,S}) - \log Z_{\max})$. At $\mu = 1$, using $\log \mathrm{vol}_\alpha(S) = -\frac{1}{2} \log\det S + c(\alpha, d)$ and dropping additive constants, $L = \overline{\mathrm{CBO}}(g_{\eta,S}) - \frac{1}{2} \log\det S$, matching PCBO_s at $s = +1$.

Proof (b). Symmetric argument with $L(\eta, S, \mu) = \log \mathrm{vol}_\alpha(S) + \mu(\underline{\mathrm{CBO}}(g_{\eta,S}) - c)$, $\mu = 1$, giving $\underline{\mathrm{CBO}}(g_{\eta,S}) - \frac{1}{2} \log\det S$, matching PCBO_s at $s = -1$.

Proof: Boundary Blow-up of the Newton Step

Framework: exact-quadratic ℓ on $\mathbb{R}^d$, so $\nabla^2\hat{\ell}$ is constant, isotropic $S = \vartheta I$.

Lemma (Boundary blow-up)

Let $\nabla^2\hat{\ell} = \sum_i \lambda_i v_i v_i^\top$ be an orthonormal eigendecomposition. If $\nabla\hat{\ell}(\eta)$ has a nonzero component along the eigenvector attaining $\lambda_{\min}$ (resp. $\lambda_{\max}$), then $\|u_s(\vartheta)\| \to \infty$ as $\vartheta \to \lambda_{\min}(\nabla^2\hat{\ell})^-$ ($s = +1$), resp. $\vartheta \to \lambda_{\max}(\nabla^2\hat{\ell})^+$ ($s = -1$).

Proof. For $s = +1$, $H_s(\vartheta) = \sum_i (\lambda_i - \vartheta) v_i v_i^\top \succ 0$ for $\vartheta < \lambda_{\min}(\nabla^2\hat{\ell})$, so

$$u_{+1}(\vartheta) = -\sum_i (\lambda_i - \vartheta)^{-1} (v_i^\top \nabla\hat{\ell}(\eta)) v_i.$$

By orthogonality, $\|u_{+1}(\vartheta)\|^2 = \sum_i (\lambda_i - \vartheta)^{-2} (v_i^\top \nabla\hat{\ell}(\eta))^2$. The term attaining $\lambda_i = \lambda_{\min}(\nabla^2\hat{\ell})$ with $v_i^\top \nabla\hat{\ell}(\eta) \neq 0$ diverges as $\vartheta \to \lambda_{\min}(\nabla^2\hat{\ell})^-$, hence $\|u_{+1}(\vartheta)\| \to \infty$. The case $s = -1$ is identical, with $H_s(\vartheta) = \sum_i (\vartheta - \lambda_i) v_i v_i^\top$, $u_{-1}(\vartheta) = -\sum_i (\vartheta - \lambda_i)^{-1} (v_i^\top \nabla\hat{\ell}(\eta)) v_i$, and $\vartheta \to \lambda_{\max}(\nabla^2\hat{\ell})^+$.

Proof: The Penalty Repairs the Degeneracy

Framework: exact-quadratic ℓ on $\mathbb{R}^d$, Assumptions (a) and (b), isotropic $S = \vartheta I$, boundary blow-up lemma above.

Proposition (Penalty breaks monotonicity)

For any $\zeta > 0$, $\mathrm{PCBO}_s(\eta, \cdot)$ has an interior stationary point: a global minimum for $s = +1$, a global maximum for $s = -1$.

Proof. For any ϑ with $H_s(\vartheta) \succ 0$, the inner objective is strictly concave ($s = +1$) or convex ($s = -1$) on $\mathbb{R}^d$, so its extremiser is unique and interior, and by Danskin's theorem, $\partial_\vartheta \mathrm{PCBO}_s = \frac{1}{2}\|u_s(\vartheta)\|^2 - \zeta/\vartheta$.

For $s = +1$, with $\lambda_{\min}(\nabla^2\hat{\ell}) > 0$ and domain $(0, \lambda_{\min}(\nabla^2\hat{\ell}))$: as $\vartheta \to 0^+$, $u_s(\vartheta) \to -[\nabla^2\hat{\ell}]^{-1}\nabla\hat{\ell}(\eta)$ stays finite, so $-\zeta \log\vartheta \to +\infty$ dominates. As $\vartheta \to \lambda_{\min}(\nabla^2\hat{\ell})^-$, the Lemma gives $\|u_s(\vartheta)\| \to \infty$, so $\mathrm{PCBO}_s \to +\infty$ there too. Extending PCBO_s by $+\infty$ at both endpoints of the compact interval $[0, \lambda_{\min}(\nabla^2\hat{\ell})]$ gives a lower semicontinuous function, hence an interior global minimum by the extreme value theorem, with $\partial_\vartheta \mathrm{PCBO}_s(\vartheta^\star) = 0$ by Fermat's theorem.

For $s = -1$, let $M := \max(0, \lambda_{\max}(\nabla^2\hat{\ell}))$, coinciding with $\lambda_{\max}(\nabla^2\hat{\ell})$ throughout this work. On domain (M, ∞): as $\vartheta \to M^+$, the Lemma gives $\|u_s(\vartheta)\| \to \infty$, so the inner infimum diverges to $-\infty$. As $\vartheta \to \infty$, $u_s(\vartheta) = O(1/\vartheta) \to 0$, so $-\zeta \log\vartheta \to -\infty$ dominates. The same argument, with $-\infty$ at both endpoints and upper semicontinuity, gives an interior global maximum.

Proof: Isotropic and Diagonal Stationarity

Framework: Assumptions (a) through (c) unchanged, (d) extended coordinatewise for the diagonal candidate.

Proposition

$$\vartheta^\star \approx \frac{d}{\|u_s\|^2}, \qquad \vartheta_j^\star \approx \frac{1}{(u_s^{\mathrm{diag}})_j^2}.$$

Proof. By Assumption (c), each inner extremiser is unique and interior, so Danskin's theorem applies. For the isotropic candidate, $\partial_\vartheta \mathrm{PCBO}_s = \frac{1}{2}\|\theta_s - \eta\|^2 - \frac{d}{2\vartheta}$. For the diagonal candidate, $\partial_{\vartheta_j} \mathrm{PCBO}_s = \frac{1}{2}(\theta_s^{\mathrm{diag}} - \eta)_j^2 - \frac{1}{2\vartheta_j}$. Setting these to zero yields the exact stationarity conditions. Substituting the second-order displacement $\theta_s - \eta \approx s\, u_s$ (with $u_s = -H_s^{-1}\nabla\hat{\ell}(\eta)$) gives $(\theta_s - \eta)_j^2 \approx (u_s)_j^2$ since $s^2 = 1$, yielding the stated approximate stationarity conditions.

Proof: Linear Convergence of the Damped Picard Map

Framework: Assumption (d), isotropic (scalar) case, extends coordinatewise to diagonal on $\prod_j \mathcal{I}_{s,j}$. Does not apply to LRD/BD, where ψ is closed-form with no iteration.

Theorem

Let $|\Phi'(\vartheta)| \le L_\Phi$ on an invariant interval $\mathcal{I}_s$, $\kappa(w) \coloneqq 1 + w(L_\Phi - 1)$. If $L_\Phi < 1$, T_w is a $\kappa(w)$-contraction for every $w \in (0, 1]$, so by Banach's fixed-point theorem it admits a unique fixed point $\vartheta^\dagger \in \mathcal{I}_s$, with $|\vartheta_t - \vartheta^\dagger| \le \kappa(w)^t |\vartheta_0 - \vartheta^\dagger|$.

Proof. Because $\mathcal{C}_s$ is a projection onto the closed convex interval $\mathcal{I}_s$, it is non-expansive, $|\mathcal{C}_s(x) - \mathcal{C}_s(y)| \le |x - y|$. By the mean value theorem, $|\Phi(\vartheta) - \Phi(\vartheta')| \le L_\Phi |\vartheta - \vartheta'|$ on $\mathcal{I}_s$. Decomposing $T_w(\vartheta) - T_w(\vartheta') = (1 - w)(\vartheta - \vartheta') + w(\mathcal{C}_s(\Phi(\vartheta)) - \mathcal{C}_s(\Phi(\vartheta')))$, the triangle inequality gives $|T_w(\vartheta) - T_w(\vartheta')| \le \kappa(w) |\vartheta - \vartheta'|$. If $L_\Phi < 1$, then $\kappa(w) < 1$, so T_w is a contraction on the complete metric space $\mathcal{I}_s$, and Banach's theorem applies.

Proof: Explicit Bound on L_Φ

Framework: isotropic (scalar) case, diagonal curvature proxy $H_s \approx s\ \mathrm{diag}(h + \delta - \vartheta)$, non-zero gradient $g \coloneqq \nabla \hat{\ell}(\eta)$.

Proposition (Explicit L_Φ)

With $a_j(\vartheta) \coloneqq h_j + \delta - \vartheta$, and setting $u_{\min}(\eta) \coloneqq \min_{\mathcal{I}_s} \|u_s\| > 0$ and $a_{\min} \coloneqq \epsilon_{\mathrm{rel}} \min_j(h_j + \delta)$, we have $|\Phi'(\vartheta)| \le \frac{2d\|g\|^2}{u_{\min}(\eta)^4 a_{\min}^3} \coloneqq L_\Phi$. This bound degrades near stationary points ($\|u_s\| \propto \|g\| \to 0$) and is uninformative at network scale.

Proof. Differentiating $\Phi(\vartheta) = d/\|u_s(\vartheta)\|^2$ gives $\Phi'(\vartheta) = -\frac{2d}{\|u_s(\vartheta)\|^4} \sum_j \frac{g_j^2}{a_j(\vartheta)^3}$. For nonzero g, taking the minimum of $\|u_s\|$ over $\mathcal{I}_s$ gives $\|u_s(\vartheta)\| \ge u_{\min}(\eta)$, and the cap keeps ϑ at least $\epsilon_{\mathrm{rel}} \min_j(h_j + \delta)$ away from the spectral boundary, so $\max_j |a_j(\vartheta)|^{-3} \le a_{\min}^{-3}$. Combining these two bounds, $|\Phi'(\vartheta)| \le \frac{2d\|g\|^2}{u_{\min}(\eta)^4} \max_j |a_j(\vartheta)|^{-3} \le \frac{2d\|g\|^2}{u_{\min}(\eta)^4 a_{\min}^3} \coloneqq L_\Phi$.

Proof: LRD and Block-Diagonal Stationarity

Framework: Assumptions (a) through (c) extended to S, (d) assumed blockwise for the block-diagonal candidate.

Proposition

Exact stationarity gives $S_\star^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top$*, rank one and singular for* $d > 1$*. Regularise to* $S^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top + \psi I$*,* $\psi > 0$*.*

Proof. By Assumption (d) extended to S, $\theta_s^{\mathrm{LRD}}(S)$ exists, is unique, lies in U, and satisfies $\theta_s^{\mathrm{LRD}}(S) \neq \eta$, so Danskin's theorem applies to $\mathrm{PCBO}_s(\eta, \cdot)$ at S, giving, with $\theta_s^{\mathrm{LRD}}(S) - \eta \approx s\, u_s^{\mathrm{LRD}}$ and $s^2 = 1$, $\nabla_S \mathrm{PCBO}_s \approx \frac{1}{2} u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top - \frac{1}{2} S^{-1}$. Setting $\nabla_S \mathrm{PCBO}_s = 0$ yields $S_\star^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top$, a rank-one matrix (since $u_s^{\mathrm{LRD}} \neq 0$ by (d)), hence singular for $d > 1$: no $S \in \mathcal{S}_{++}^d$ solves the stationarity condition exactly. Adding ψI gives $S^{-1} = S_\star^{-1} + \psi I$, the sum of a positive semi-definite and a strictly positive definite matrix, hence positive definite: for every $v \neq 0$, $v^\top S^{-1} v \geq \psi \|v\|^2 > 0$. BD applies this per block.

Lemma: Spectrum of the LRD Regulariser

Framework: LRD precision $S^{-1} = uu^\top + \psi I$ with $\psi > 0$ and $u \neq 0$ (writing $u := u_s^{\mathrm{LRD}}$ for brevity).

Lemma

S^{-1} has eigenvalue $\psi + \|u\|^2$ on $\mathrm{span}\{u\}$ and ψ on $u^\perp$. Hence S has reciprocal eigenvalues on the same eigenspaces:

$$\lambda_{\min}(S) = \frac{1}{\psi + \|u\|^2}.$$

Proof. For $v = cu$ ($c \in \mathbb{R}$),

$$S^{-1}v = (uu^\top + \psi I)(cu) = c\|u\|^2 u + c\psi u = (\|u\|^2 + \psi)v :$$

eigenvalue $\psi + \|u\|^2$ along u. For $v \perp u$, $u^\top v = 0$, so

$$S^{-1}v = (uu^\top + \psi I)v = u(u^\top v) + \psi v = \psi v :$$

eigenvalue ψ on $u^\perp$ ($d - 1$ directions). Since $S = (S^{-1})^{-1}$ shares the same eigenvectors with reciprocal eigenvalues, the largest eigenvalue of S^{-1} ($\psi + \|u\|^2 > \psi$, as $\|u\|^2 > 0$) becomes the *smallest* eigenvalue of S: $\lambda_{\min}(S) = 1/(\psi + \|u\|^2)$.

Proof: LRD Feasibility for $s = -1$

Framework: LRD precision $S^{-1} = u_s^{\mathrm{LRD}}(u_s^{\mathrm{LRD}})^\top + \psi I$, diagonal curvature proxy $H_s \approx S - \mathrm{diag}(h + \delta)$ under $s = -1$, conditional, unlike $s = +1$, which is unconditional.

Theorem

$H_s \succ 0 \iff S \succ \mathrm{diag}(h + \delta)$. A sufficient condition is $\lambda_{\min}(S) > M$, equivalently $\psi + \|u_s^{\mathrm{LRD}}\|^2 < 1/M$ (M: max curvature). Setting $\psi = \frac{1}{\varphi}(\frac{1}{M} - \|u_s^{\mathrm{LRD}}\|^2)$, $\varphi > 1$, certifies this whenever $\|u_s^{\mathrm{LRD}}\|^2 < 1/M$.

Proof. By the spectrum lemma (previous slide), $\lambda_{\min}(S) = 1/(\psi + \|u_s^{\mathrm{LRD}}\|^2)$, so $\lambda_{\min}(S) > M \iff \psi < 1/M - \|u_s^{\mathrm{LRD}}\|^2$. If this holds, $S \succeq \lambda_{\min}(S) I \succ MI \succeq \mathrm{diag}(h + \delta)$, giving $H_s \succ 0$ under the proxy. This upper bound is strictly positive if and only if $\|u_s^{\mathrm{LRD}}\|^2 < 1/M$. Under this condition, $\psi = \varphi^{-1}(1/M - \|u_s^{\mathrm{LRD}}\|^2)$ guarantees both $H_s \succ 0$ and $\psi > 0$ for any $\varphi > 1$.

Proof: The Indicator Candidate Is Exactly SAM

Framework: $\Theta \subseteq \mathbb{R}^d$ closed, ℓ continuous, $\pi \equiv 1_\Theta$ the uninformative prior, $g_{\eta,\rho}(\theta) := 1[\|\theta - \eta\|_2 \le \rho]$.

Proposition (12)

$g_{\eta,\rho} \in \mathcal{F}(\Theta)$ and $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\sup_{\theta \in \Theta \cap \bar{B}(\eta,\rho)} \ell(\theta)$, so
$\arg\max_\eta \underline{\mathrm{CBO}}(g_{\eta,\rho}) = \arg\min_\eta \mathcal{E}_{\mathrm{SAM}}(\eta; \rho, 0)$. *If $\nabla \ell(\eta) \neq 0$ and $B(\eta, \rho) \subseteq \Theta$,*
$\varepsilon^\star_{\mathrm{SAM}} = \rho \nabla \ell(\eta) / \|\nabla \ell(\eta)\|_2$.

Proof. $g_{\eta,\rho}(\eta) = 1$ gives $\sup_\theta g_{\eta,\rho} = 1$, so $g_{\eta,\rho} \in \mathcal{F}(\Theta)$. Off $\Theta \cap \bar{B}(\eta, \rho)$, $\log g_{\eta,\rho} = -\infty$, so the inner term is $+\infty$, excluded from the infimum since it is finite on $\Theta \cap \bar{B}(\eta, \rho)$ (nonempty, contains η). On $\Theta \cap \bar{B}(\eta, \rho)$ the term is $-\ell(\theta)$, giving $\underline{\mathrm{CBO}}(g_{\eta,\rho}) = -\sup_{\Theta \cap \bar{B}(\eta,\rho)} \ell$, attained since $\Theta \cap \bar{B}(\eta, \rho)$ is compact and ℓ continuous. Substituting $\theta = \eta + \varepsilon$ and negating gives the objective identity.

For the perturbation: under $B(\eta, \rho) \subseteq \Theta$ the constraint reduces to $\|\varepsilon\|_2 \le \rho$, and the linear model $\ell(\eta) + \nabla \ell(\eta)^\top \varepsilon$ is maximised, by Cauchy–Schwarz, uniquely at ε a nonnegative multiple of $\nabla \ell(\eta)$ with $\|\varepsilon\|_2 = \rho$.

SAM–Bayes: The Probabilistic Analog

Framework how the SAM–Bayes link works on the probabilistic side, for comparison with the exact possibilistic identity of the indicator candidate.

SAM as a relaxation of Bayes [14]

Möllenhoff and Khan (ICLR 2023) show SAM is a **relaxation** of Bayes, not the Bayes objective itself. Bayes maximizes an expected negative-loss under a posterior q, SAM minimizes a worst-case loss over a neighbourhood, max-loss versus expected-loss. Their connection replaces the expected-loss term by its **optimal convex lower bound**, via the Fenchel biconjugate. That relaxed objective, for an isotropic Gaussian posterior, recovers exactly SAM's update, the posterior mean matching SAM's iterate and the fixed variance controlling the neighbourhood radius.

Contrast with the possibilistic case

The probabilistic link goes through a **bound**, relaxed-Bayes always lower-bounds the true Bayes objective, an approximation not an identity. On the possibilistic side, with the *indicator* candidate, the lower CBO is **exactly** $-\mathcal{E}_{\mathrm{SAM}}$, no relaxation gap, though exactness is lost once the candidate is softened to a Gaussian.

Proof: Gaussian Lower CBO Relaxes the Hard Ball

Framework: $\Theta = \mathbb{R}^d$, $\pi \equiv 1_\Theta$, $g_{\eta,\vartheta}(\theta) = \mathcal{N}^*(\theta; \eta, \vartheta^{-1} I)$.

Proposition (12)

$$\underline{\mathrm{CBO}}(g_{\eta,\vartheta}) = -\sup_{\theta \in \mathbb{R}^d} \left\{ \ell(\theta) - \tfrac{\vartheta}{2} \|\theta - \eta\|_2^2 \right\}$$

If $\nabla \ell$ is globally L-Lipschitz with $L < \vartheta$, the maximiser $u^\star$ of $u \mapsto \ell(\eta + u) - \frac{\vartheta}{2}\|u\|_2^2$ exists and is unique. Under the first-order local model, its maximiser is $u^\star_{\mathrm{soft}} = \vartheta^{-1} \nabla \ell(\eta)$.

Proof. Substituting $-\log g_{\eta,\vartheta} = \frac{\vartheta}{2}\|\theta - \eta\|_2^2$ into $\underline{\mathrm{CBO}}$ gives $-\sup_\theta \{\ell(\theta) - \frac{\vartheta}{2}\|\theta - \eta\|_2^2\}$. Set $\theta = \eta + u$, $\phi(u) := \ell(\eta + u) - \frac{\vartheta}{2}\|u\|_2^2$. The descent lemma from L-Lipschitz $\nabla \ell$ gives ϕ $(\vartheta - L)$-strongly concave for $\vartheta > L$, hence coercive with a unique global maximiser $u^\star$. Under the first-order model, the objective becomes $\ell(\eta) + \nabla \ell(\eta)^\top u - \frac{\vartheta}{2}\|u\|_2^2$, whose gradient vanishes uniquely at $u^\star_{\mathrm{soft}} = \vartheta^{-1} \nabla \ell(\eta)$, with Hessian $-\vartheta I \prec 0$. This coincides with $u^\star$ when ℓ is affine near η, since the approximation is then exact.

Derivation: Adaptive SAM Radius from the Learned Precision

Framework: level-set bridge at $\alpha = e^{-1/2}$, combined with the isotropic stationarity result (Proof: Isotropic and Diagonal Stationarity) at $s = -1$.

Superlevel-set bridge

For $g_{\eta,\vartheta} = \mathcal{N}^*(\theta; \eta, \vartheta^{-1} I)$, the superlevel set at any $\alpha \in (0, 1)$ is a closed Euclidean ball:

$$\{\theta : g_{\eta,\vartheta}(\theta) \geq \alpha\} = \bar{B}(\eta, \rho_\alpha), \qquad \rho_\alpha \coloneqq \vartheta^{-1/2} \sqrt{-2 \log \alpha}.$$

Fixing $\alpha = e^{-1/2}$ gives $\rho_\alpha = \vartheta^{-1/2}$.

Derivation. Reusing $\vartheta^\star = d / \|u_{-1}\|_2^2$ with Newton step $u_{-1} \approx -H_{-1}^{-1} \nabla \hat{\ell}(\eta)$ (isotropic stationarity), the induced radius $\rho_\alpha = \vartheta^{-1/2}$ becomes

$$\rho^\star(\eta) \approx \frac{1}{\sqrt{d}} \left\| H_{-1}^{-1} \nabla \hat{\ell}(\eta) \right\|_2.$$

No separate optimiser is needed: the isotropic update at $s = -1$ already produces $\vartheta^\star$, from which $\rho^\star(\eta)$ is immediate. Whether this relaxation is theoretically sound (trading the exact indicator identity for a Gaussian approximation) is left open, and tested experimentally.

Algorithms: CBOpt-iso and CBOpt-diag

CBOpt-iso

Require: d, $s \in \{\pm 1\}$, $\{\rho_t\}$, $\delta > 0$, $r, \epsilon_{\mathrm{rel}} > 0$, $\beta_{\mathbf{1}}, \beta_{\mathbf{2}} \in [0,1)$, $h_{\mathbf{0}} > 0$, $w \in (0,1]$, $K \geq 1$

$\eta \leftarrow$ weights, $h \leftarrow h_{\mathbf{0}}\mathbf{1}$, $g \leftarrow 0$, $\vartheta \leftarrow (h_{\mathbf{0}} + \delta)(1 - s\epsilon_{\mathrm{rel}})$
for $t = 1, 2, \ldots$ **do**
  $\hat{g} \leftarrow \widehat{\nabla}\ell(\eta)$, $g \leftarrow \beta_{\mathbf{1}} g + (1 - \beta_{\mathbf{1}})\hat{g}$,
  $h \leftarrow \beta_{\mathbf{2}} h + (1 - \beta_{\mathbf{2}})\hat{g}^{\odot \mathbf{2}}$, $\bar{g} \leftarrow g/(1 - \beta_{\mathbf{1}}^t)$
  $m \leftarrow \min_j(h_j + \delta)$, $M \leftarrow \max_j(h_j + \delta)$
  for $k = 1, \ldots, K$ **do**
    $u \leftarrow -s(\mathrm{diag}(h + \delta) - \vartheta I)^{-\mathbf{1}}(\bar{g} + \delta\eta)$
    $\vartheta^\star \leftarrow d/\|u\|^{\mathbf{2}}$
    $\vartheta_{\mathrm{c}}^\star \leftarrow \mathcal{C}_s(\vartheta^\star)$
    $\vartheta \leftarrow (1 - w)\vartheta + w\vartheta_{\mathrm{c}}^\star$
  end for
  $\eta \leftarrow \eta + \rho_t u$
end for
return η, ϑ

CBOpt-diag

Require: d, $s \in \{\pm 1\}$, $\{\rho_t\}$, $\delta > 0$, $r, \epsilon_{\mathrm{rel}} > 0$, $\beta_{\mathbf{1}}, \beta_{\mathbf{2}} \in [0,1)$, $h_{\mathbf{0}} > 0$, $w \in (0,1]$, $K \geq 1$

$\eta \leftarrow$ weights, $h \leftarrow h_{\mathbf{0}}\mathbf{1}$, $g \leftarrow 0$, $\boldsymbol{\vartheta} \leftarrow (h_{\mathbf{0}} + \delta)(1 - s\epsilon_{\mathrm{rel}})\mathbf{1}$
for $t = 1, 2, \ldots$ **do**
  update $g, h, \bar{g}$ **as** CBOpt-iso
  for $k = 1, \ldots, K$ **do**
    $u \leftarrow -s(h + \delta - \boldsymbol{\vartheta})^{-\mathbf{1}} \odot (\bar{g} + \delta\eta)$
    $\boldsymbol{\vartheta}^\star \leftarrow 1/(u \odot u)$
    $\boldsymbol{\vartheta}_{\mathrm{c}}^\star \leftarrow \mathcal{C}_s^{\mathrm{diag}}(\boldsymbol{\vartheta}^\star)$
    $\boldsymbol{\vartheta} \leftarrow (1 - w)\boldsymbol{\vartheta} + w\boldsymbol{\vartheta}_{\mathrm{c}}^\star$
  end for
  $\eta \leftarrow \eta + \rho_t u$
end for
return $\eta, \boldsymbol{\vartheta}$

Cost: $O(Kd)$ per mode step (K damped Picard iterations, each $O(d)$ since H_s is diagonal).

Algorithms: CBOpt-LRD and CBOpt-BD

CBOpt-LRD

Require: $s \in \{\pm 1\}$, $\{\rho_t\}$, $\delta > 0$, $a, r > 0$, $\xi \in (0,1)$, $\varphi > 1$, $\beta_{\mathbf{1}}, \beta_{\mathbf{2}} \in [0,1)$, $h_{\mathbf{0}} > 0$

$\eta \leftarrow$ weights, $h \leftarrow h_{\mathbf{0}}\mathbf{1}$, $g \leftarrow 0$, $S \leftarrow \delta I$, $s_{\mathrm{e}} \leftarrow s$
for $t = 1, 2, \ldots$ **do**
  update $g, h, \bar{g}$ as CBOpt-iso
  $m \leftarrow \min_j(h_j + \delta)$, $M \leftarrow \max_j(h_j + \delta)$
  $u_{\max}^{\mathbf{2}} \leftarrow (1-\xi)/M$
  $u \leftarrow -s_{\mathrm{e}}(\mathrm{diag}(h+\delta) - S)^{-\mathbf{1}}(\bar{g} + \delta\eta)$
  if $s = -\mathbf{1} \wedge \|u\|^{\mathbf{2}} \le u_{\max}^{\mathbf{2}}$ **then** $s_{\mathrm{e}} \leftarrow -\mathbf{1}$
  else if $s = -\mathbf{1}$ **then** $s_{\mathrm{e}} \leftarrow +\mathbf{1}$
  end if
   if $s_{\mathrm{e}} = +\mathbf{1}$ **then** $\psi \leftarrow \max(r, 1/m + a)$
   else $\psi \leftarrow \max(r, \varphi^{-\mathbf{1}}(1/M - \|u\|^{\mathbf{2}}))$
   end if
   $S \leftarrow (uu^{\top} + \psi I)^{-\mathbf{1}}$, $\eta \leftarrow \eta + \rho_t u$
end for
return η, S

CBOpt-BD

Require: blocks $\ell \in \{1, \ldots, L\}$, $s \in \{\pm 1\}$, $\{\rho_t\}$, $\delta > 0$, $a, r > 0$, $\xi \in (0,1)$, $\varphi > 1$, $\beta_{\mathbf{1}}, \beta_{\mathbf{2}} \in [0,1)$, $h_{\mathbf{0}} > 0$

$\eta \leftarrow$ weights, $h \leftarrow h_{\mathbf{0}}\mathbf{1}$, $g \leftarrow 0$, $S_\ell \leftarrow \delta I_{d_\ell}$, $s_{\mathrm{e},\ell} \leftarrow s$
for $t = 1, 2, \ldots$ **do**
  update $g, h, \bar{g}$ as CBOpt-iso
  for $\ell = 1, \ldots, L$ **do**
    $m_\ell, M_\ell \leftarrow \min_{j\in\ell}, \max_{j\in\ell}(h_j + \delta)$
    $u_{\max,\ell}^{\mathbf{2}} \leftarrow (1-\xi)/M_\ell$
    $u_\ell \leftarrow -s_{\mathrm{e},\ell}(\mathrm{diag}(h_\ell + \delta) - S_\ell)^{-\mathbf{1}}(\bar{g}_\ell + \delta\eta_\ell)$
    flip $s_{\mathrm{e},\ell}$ as CBOpt-LRD, per block
    ψ_ℓ as CBOpt-LRD, per block
     $S_\ell \leftarrow (u_\ell u_\ell^{\top} + \psi_\ell I)^{-\mathbf{1}}$
  end for
  $\eta \leftarrow \eta + \rho_t(u_{\mathbf{1}}, \ldots, u_L)$
end for
return $\eta, \{S_\ell\}$

Cost: $O(d)$ per mode step (Sherman–Morrison, no Picard iteration; ψ closed-form).

Reminder: The Sherman–Morrison Formula

Sherman–Morrison identity

For $A \in \mathbb{R}^{d \times d}$ invertible and $u, v \in \mathbb{R}^d$, if $1 + v^\top A^{-1} u \neq 0$, then $A + uv^\top$ is invertible with

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1} uv^\top A^{-1}}{1 + v^\top A^{-1} u}.$$

- A rank-one update to A produces a rank-one update to its inverse: no need to invert the updated matrix directly
- Naive inversion of a dense $d \times d$ matrix (Gaussian elimination or LU) costs $O(d^3)$
- Applied here with $A = \psi I$ (diagonal, trivially invertible) and $u = v$: inverting $S^{-1} = uu^\top + \psi I$ to get S costs $O(d)$ instead

Proof: Storage and Inversion Complexity

Framework: The main argument targets the LRD form $S^{-1} = uu^\top + \psi I$ (BD applies it per block). Isotropic and diagonal are simpler, with H_s already diagonal.

Proposition

Per mode step, storing the precision costs $O(1)$ for isotropic and $O(d)$ for diagonal, LRD, BD ($\sum_\ell O(d_\ell) = O(d)$). Computing $H_s^{-1}(\bar{g}_t + \delta\eta_t)$ once costs $O(d)$ for all four.

Proof. Storage is immediate: a scalar ϑ ($O(1)$) for isotropic, $\boldsymbol{\vartheta} \in \mathbb{R}^d$ ($O(d)$) for diagonal, and, by the spectrum of the LRD regulariser, $S^{-1} = uu^\top + \psi I$ stored as $(u, \psi) \in \mathbb{R}^d \times \mathbb{R}_{>0}$ ($O(d)$) for LRD, summed over L blocks for BD.

For inversion, $S = \frac{1}{\psi} I - b\, uu^\top$ with $b = 1/(\psi(\psi + \|u\|^2))$. Writing $P = \mathrm{diag}(h + \delta) - \frac{1}{\psi} I$, strictly invertible since $1/\psi < h_j + \delta$ ($s = +1$) or $1/\psi > h_j + \delta$ ($s = -1$), $H_s = s(P + b\, uu^\top)$. Sherman–Morrison gives, with $z := \bar{g}_t + \delta\eta_t$,

$$H_s^{-1} z = s\big(P^{-1} z - b(P^{-1}u)(u^\top P^{-1} z)/(1 + b\, u^\top P^{-1} u)\big).$$

The denominator never vanishes, since the safeguard theorems guarantee $H_s \succ 0$, so $sH_s = P + b\, uu^\top$ is invertible, and by the matrix determinant lemma $\det(P)(1 + bu^\top P^{-1} u) = \det(P + buu^\top) \neq 0$. Diagonal P makes every operation coordinatewise, costing $O(d)$ with no dense $d \times d$ matrix formed. BD applies the same solve per block; isotropic and diagonal inversions are directly $O(d)$, H_s being diagonal in both cases.

Detail: Toy Problem Dimension Sweep

Diagonal quadratic loss, $d \in [3.5 \times 10^3, 1.9 \times 10^5]$, mean ms per step over 6 runs.

d	LRD	BD	iso	diag
3,466	1.47 ± 0.04	2.30 ± 0.12	3.27 ± 0.17	3.86 ± 0.32
8,970	1.63 ± 0.18	2.33 ± 0.15	3.24 ± 0.11	4.20 ± 0.31
26,122	2.40 ± 0.11	3.63 ± 0.21	3.88 ± 0.34	7.13 ± 0.27
85,002	2.75 ± 0.20	3.50 ± 0.06	$\mathbf{8.07 \pm 0.55}$	$\mathbf{7.96 \pm 2.01}$
190,410	5.05 ± 0.50	5.28 ± 0.24	10.05 ± 0.71	12.01 ± 0.42

Table. Runtime scaling with dimension, for each precision structure.

- **Ordering**: $\mathrm{LRD} < \mathrm{BD} < \mathrm{iso} < \mathrm{diag}$ holds at 4 of 5 dimensions, matching the predicted $O(d)$ vs $O(Kd)$ split
- **Factor**: $\sim 2\times$, not $5\times$ ($K = 5$), and the internal ordering (LRD vs BD's L per-block solves, iso vs diag's elementwise step) both plausibly reflect GPU kernel-launch overhead
- **Swap**: at $d = 85{,}002$, iso and diag swap, but their ranges overlap (8.07 ± 0.55 vs 7.96 ± 2.01)

Ablation: Sensitivity to the Penalty Coefficient ζ

Isotropic toy problem, scaling the penalty coefficient over $(\frac{1}{4}, \frac{1}{2}, 1, 2, 4)$, a $16\times$ range.

- $s = +1$: $\vartheta^\star$ moves $0.074 \to 0.21$ $(2.8\times)$, sub-linear in the $16\times$ range
- $s = -1$: $\vartheta^\star$ moves $55 \to 8.5$ $(6.4\times)$, more sensitive

Proposition

For $S = \vartheta I$, $-\frac{1}{2}\log\det S = -\frac{d}{2}\log\vartheta$. $-\zeta\log\vartheta$ matches the matrix penalty's isotropic restriction ***only for*** *$\zeta = d/2$.*

$\vartheta^\star$ is not robust to arbitrary scaling of the coefficient, especially for $s = -1$. Fixing $\zeta = d/2$ via the volume argument removes this free parameter entirely, for both signs.

Reminder: Evaluation Metrics

Framework: checkpoint at lowest validation loss. In-domain metrics on Fashion-MNIST. OOD metrics on Fashion-MNIST vs EMNIST-Letters using max-softmax confidence $\max_k \mathrm{softmax}(\mathrm{logits})_k$.

- **Accuracy**: fraction of correctly classified test examples
- **NLL**: with $p_k(x) := \mathrm{softmax}(\mathrm{logits}(x))_k$ and y_i the true label,

$$\mathrm{NLL} = -\frac{1}{n}\sum_{i=1}^{n} \log p_{y_i}(x_i)$$

  Penalises a confidently wrong prediction far more than an unsure one
- **ECE**: over 15 equal-width confidence bins $B_1, \ldots, B_{15}$,

$$\mathrm{ECE} = \sum_{m=1}^{15} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|$$

  Measures discrepancies between stated confidence and actual accuracy
- **FPR@95**: mean $\mathrm{FPR}(\tau)$ over `roc_curve` points with $\mathrm{TPR}(\tau) \leq 0.9505$. Penalises unfamiliar inputs assigned high confidence
- **AUROC**: area under the ROC curve, true vs false positive rate across all thresholds τ. Quantifies overall separation between in-distribution and OOD confidence

Results: Full Reproduction Table

Optimiser	Acc. (%)	NLL	ECE	FPR@95	AUROC
`uCBOpt`	90.9 ± 0.2	0.272 ± 0.004	0.025 ± 0.001	0.263 ± 0.005	0.795 ± 0.005
IVON	**91.0±0.2**	**0.258±0.003**	**0.019±0.001**	**0.241±0.009**	**0.812±0.012**
AdamW	89.6 ± 0.3	0.296 ± 0.005	0.022 ± 0.006	0.335 ± 0.005	0.722 ± 0.016
Reference values [1]					
`uCBOpt`	90.7 ± 0.2	0.270 ± 0.002	0.023 ± 0.000	**0.245±0.010**	**0.810±0.012**
IVON	**91.1±0.3**	**0.258±0.001**	**0.013±0.004**	0.249 ± 0.024	0.804 ± 0.030
AdamW	90.0 ± 0.5	0.294 ± 0.007	0.026 ± 0.004	0.321 ± 0.008	0.745 ± 0.009

Table. LeNet-5, Fashion-MNIST (in-domain) vs EMNIST-Letters (OOD), 100 epochs, 3 seeds. All five metrics, own runs versus reference values.

- **Every metric/optimiser pair** matches the reference within 2σ
- **OOD ranking flips** (`uCBOpt` vs IVON on FPR@95/AUROC), but within IVON's own noise

Complete Hyperparameters (1/2): Baselines and Data

Compute: Google Colab, T4 GPUs.

Architecture: LeNet-5 variant ($d = 44{,}426$), two conv layers ($1 \to 6$, $6 \to 16$, 5×5 kernels) with 2×2 max-pooling and ReLU, three FC layers ($256 \to 120 \to 84 \to 10$).

Data: Fashion-MNIST, 60,000 training images split 90/10 (54,000 train / 6,000 validation), plus 10,000 standard test images (70,000 total, batch 128, no augmentation). OOD set: EMNIST-Letters, 14,800 test images.

Training: 100 epochs, linear warmup (5 epochs) then cosine decay to zero, no early stopping, checkpoint at lowest validation loss, 3 seeds $(0, 1, 2)$, $\beta_1 = 0.9$ for all optimisers.

Optimiser	ρ_t (lr)	δ	β_2	h_0	other
AdamW	10^{-3}	10^{-2}	–	–	–
IVON	0.2	2×10^{-3}	0.99999	0.5	ESS 5×10^4, 1 MC sample/step
fixed `uCBOpt`	10^{-2}	2×10^{-3}	–	0.05	$\vartheta_0 = 8 \times 10^{-6}$

Complete Hyperparameters (2/2): Learned-Precision Variants

All share $\beta_1 = 0.9$, $\beta_2 = 0.99999$. `CBOpt-iso`/`CBOpt-diag` use $K = 5$ damped Picard steps, damping $w = 0.5$; `CBOpt-LRD`/`CBOpt-BD` compute ψ in closed form.

Group	Optimiser	ρ_t (lr)	δ	h_0	other
Benchmark ($s = +1$)	`uCBOpt-iso/-diag`	10^{-2}	2×10^{-3}	0.05	$\epsilon_{\mathrm{rel}} = 0.2$, $r = 10^{-2}$
	`uCBOpt-BD`	10^{-2}	2×10^{-3}	0.05	$a = 1$, 1 block/tensor
Geometry check ($s = -1$)	`lCBOpt-LRD/-BD`	2×10^{-3}	2×10^{-3}	0.1	$a = 1$, $\xi = 0.1$, $\varphi = 2$
SAM ($s = -1$)	`lCBOpt-iso`	2×10^{-3}	2×10^{-3}	0.1	$\epsilon_{\mathrm{rel}} = 0.2$, $r \in \{0.500, 0.475, 0.525\}$
	standard SAM (SGD)	5×10^{-3}	–	–	momentum 0.9, w.d. 10^{-5}, $\rho \in \{1, 0.95, 1.05\} \times \bar{\rho}_{\mathrm{nom}}$

Diagnostic: Training and Validation Curves

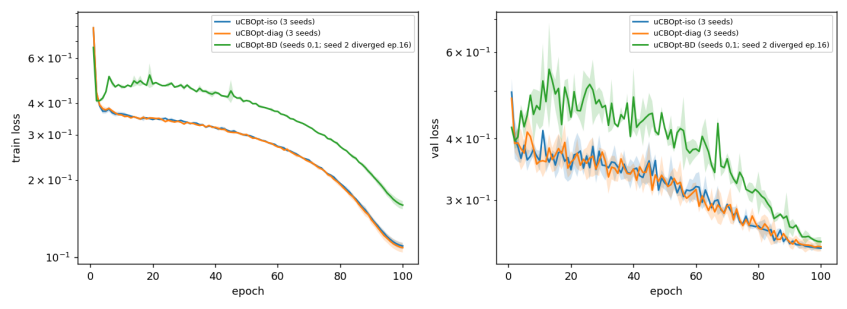


Figure. Training and validation loss, mean over seeds, LeNet-5 and Fashion-MNIST, 100 epochs.

- **Training gap**: `uCBOpt-BD` has higher training loss (structural), but closely matches the others on validation loss, so this gap does not hurt generalisation

Diagnostic: The uCBOpt-BD Divergence on Seed 2

- One of three benchmark seeds diverged at epoch 16 (training loss $0.47 \to 1.9 \times 10^7$)
- Sherman–Morrison structure: diagonal part $P_\ell = \mathrm{diag}(h_\ell + \delta) - \frac{1}{\psi_\ell} I_{d_\ell}$ has smallest entry $\approx a\, m_\ell^2$ under $\psi_\ell = 1/m_\ell + a$, amplifying ill-conditioned directions by $\approx 1/(a\, m_\ell^2) \approx 800$, identical across seeds
- Fraction of blocks with $\|u_\ell\|^2 \geq 1/M_\ell$ jumps $0.52 \to 0.82 \to 1.00$ over epochs 15 to 17 on seed 2 (vs 0.52 to 0.67 on the other two), a stochastic gradient spike, not an algebraic flaw
- No step-norm clipping or damping used, since the LRD regularisation already certifies $H_s \succ 0$ analytically

Positivity Checks: Methodology and Feasibility

Positive-definiteness check

For `CBOpt-iso`/`CBOpt-diag`, H_s is diagonal: $H_s \succ 0$ reduces to checking, coordinatewise, the sign of $s(h_j + \delta - \vartheta)$ at the uncapped target $\vartheta^\star = d/\|u\|^2$ (isotropic) or $\vartheta_j^\star = 1/(u_j)^2$ (diagonal). For `CBOpt-LRD`/`CBOpt-BD`, what is tracked is the feasibility threshold $\|u_\ell\|^2 \geq 1/M_\ell$, which governs whether a regulariser ψ_ℓ certifying $H_s \succ 0$ exists. The run flips to $s_{\mathrm{e}} = +1$ once $\|u_\ell\|^2$ exceeds the margin $(1-\xi)/M_\ell < 1/M_\ell$.

Non-emptiness of `lCBOpt-iso`'s capping interval

For $s = -1$, the invariant interval $\mathcal{I}_{-1} = \left[\max_j(h_j + \delta)(1+\epsilon_{\mathrm{rel}}),\ \max_j(h_j+\delta)/r\right]$ is non-empty iff $r < 1/(1+\epsilon_{\mathrm{rel}})$, i.e. $r < 0.833$ at $\epsilon_{\mathrm{rel}} = 0.2$. Centring the SAM sweep on $r_{\mathrm{nom}} = 0.50$ keeps $r \in [0.475, 0.525]$ well inside this range, so the cap $\max_j(h_j+\delta)/r$ stays the active bound throughout the sweep and ϑ depends strictly on r.

Code Repositories

Three repositories accompany the report, one per contribution:

- **Reproduction** (Section: reproducing `uCBOpt` at fixed precision)
  `https://github.com/mariadass-gabriel/MDV_reprod`
- **Landscapes** (Section: landscapes on tractable posteriors)
  `https://github.com/mariadass-gabriel/MDV_lands`
- **Learned precision** (Sections: constructing and applying the learned precision)
  `https://github.com/mariadass-gabriel/MDV_spread`

Each repository contains the code and logs behind every number reported.

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